

Automation Risk, Redistribution, and Economic Vulnerability: Evidence from the European Social Survey, 2002–2022

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September 12, 2026

Abstract

This paper asks whether occupational exposure to automation is associated with support for government redistribution, and when that association is strongest. We reconstruct and extend the research design of Thewissen and Rueda [2019] with all eleven European Social Survey (ESS) waves from 2002 to 2022, covering more than 275,000 respondents in 39 countries. A formal insurance model derives the hypotheses as comparative statics; its institutional predictions depend on whether income after displacement remains proportional to prior earnings. The positive association between automation risk and redistribution preferences reappears in the 2002–2012 period but is smaller in the subsequent decade. Country-by-year estimates show substantial variation across national settings. We then examine two mechanisms. Automation exposure is associated with perceived economic vulnerability, including financial strain and job insecurity. Institutions and industrial relations condition the association between exposure and perceived insecurity: after adjustment for business-cycle dynamics, union density weakens it within and between countries, whereas net income inequality and strict employment protection strengthen it. The latter pattern is consistent with insider–outsider dualization.

1 Introduction

The spread of automation and artificial intelligence has renewed debate over the political consequences of labor-market displacement. Comparative political economists use public-opinion data to study how economic vulnerability shapes policy preferences. In the benchmark study for this literature, Thewissen and Rueda [2019] found that respondents in occupations with greater routine-task automation risk expressed more support for redistribution. This paper asks whether that association persists across periods and national contexts.

We rebuild the Thewissen–Rueda design from raw ESS records harmonized to a common format for eleven waves between 2002 and 2022. First, we extend the temporal coverage beyond 2012, when cognitive automation and artificial intelligence became more salient. Second, we estimate country and country-year slopes to assess whether pooled estimates conceal geographic variation. Third, we examine whether perceived vulnerability links occupational exposure to redistribution preferences and whether welfare-state and labor-market institutions condition that link. Fourth, we derive the hypotheses from a formal insurance model. The model gives the preferred tax rate in closed form and nests the Meltzer and Richard [1981] and Moene and Wallerstein [2001] cases at opposite ends of the risk-aversion range. Its institutional comparative statics depend on a single condition: whether income in the bad state is proportional to prior earnings. Relative to the existing literature, the paper adds the first replication and extension of the benchmark design across the full ESS panel, together with a formal model that states the exact conditions behind the benchmark income-by-risk interaction. It also adds estimates that separate within-country from between-country institutional moderation.

Methodologically, the empirical literature operationalizes automation risk through occupational exposure measures, most often the routine-task and engineering-bottleneck probability of Frey and Osborne [2017] or the routine-task-intensity index of Goos et al. [2014]. Both measures capture the same underlying idea: codifiable, repetitive tasks are the ones most readily displaced by digitization and machine automation. We use both throughout, treating the Frey–Osborne probability as the reference measure and the routine-task-intensity index as the closest robustness analogue to the Thewissen–Rueda benchmark.

The findings are as follows. The positive association between automation risk and redistribution preferences reappears in the 2002–2012 window and brackets the published benchmark. After 2014 the association weakens under every roster of countries in the analysis, and variation across countries exceeds variation over time. Automation exposure predicts financial strain, expected job loss, and perceived job insecurity in every wave with the relevant items. Once business-cycle dynamics enter the models, union density weakens the exposure–strain link within and between countries, while net income inequality and strict employment protection strengthen it. Design-weighted routine exposure itself declined across rounds for the benchmark roster.

We first report the Frey–Osborne replication of the Thewissen–Rueda design and its post-2014 attenuation. We then examine country-year slopes and composition-adjusted yearly estimates. The final section presents the vulnerability and institutional analyses; routine-task-intensity (RTI) results and additional robustness checks appear in the appendix. The theory section closes with a formal insurance model whose comparative statics yield the hypotheses; derivations appear in Appendix 8.1.

2 Theory

2.1 Comparative Political Economy of Redistribution

Our framework draws on two accounts of welfare-state development and redistribution. Wilensky’s modernization account treats welfare-state expansion as a response to industrialization, economic growth, and demographic change [Wilensky, 1974]. Stephens’ power-resources account emphasizes labor organization, socialist parties, and collective bargaining as forces that sustain redistribution [Stephens, 1979]. Together, these accounts distinguish the creation of economic risk from its political consequences: institutions and class power shape how people experience new risks.

The comparative political economy literature also clarifies why redistribution demand need not mechanically follow from objective exposure. Moene and Wallerstein [2001] show that preferences over social insurance and redistribution depend on the distribution of risk and income,

not on inequality alone. Iversen and Soskice’s asset theory similarly treats social-policy preferences as a function of exposure to labor-market risk and the portability of workers’ skills [Iversen and Soskice, 2001]. Korpi and Palme’s paradox of redistribution adds an institutional feedback: encompassing welfare states can reduce poverty and inequality because broad social programs build larger political coalitions for redistribution [Korpi and Palme, 1998]. These arguments imply that automation exposure should matter politically when it is experienced as uninsured labor-market risk, and that welfare-state and labor-market institutions should condition that experience.

2.2 Job Automation and Economic Threat

Automation can alter perceptions of economic threat, which in turn can shape political attitudes. Research on economic shocks links perceived insecurity, especially anticipated unemployment, to changes in political and social preferences [Margalit, 2019, 2013]. Research on economic vulnerability and political psychology likewise treats perceived threats as a potential source of support for protective policies [Hacker et al., 2013, Rogers, 1975]. Automation is one potential source of such disruption: people in exposed occupations may regard their employment prospects as less secure [Autor and Salomons, 2018].

2.3 Automation Risk as Conditional Redistribution Demand

The benchmark claim is that routine-task automation risk increases demand for redistribution because occupational exposure can signal future income and employment losses [Thewissen and Rueda, 2019]. Respondents in routine, automatable occupations may anticipate displacement or downward mobility, making redistribution and social insurance more attractive. We therefore expect automation exposure to be positively associated, on average, with support for redistribution.

The association need not be constant over time. Economic development, employment structures, and institutions change, altering how occupational risk is understood. Recessions, recoveries, and inflation can make the same exposure feel more or less threatening. People and firms also adapt as automation diffuses, through changes in expectations, tasks, and occupations.

Routine-task measures may therefore lose political relevance as technological change advances beyond routine-biased computerization. We expect the automation–redistribution association to vary across survey waves rather than to move monotonically in one direction.

The same exposure may also have different consequences across countries. Welfare states and labor-market institutions shape whether automation risk is privately borne or collectively insured. Strong labor organization and broad bargaining can give workers power before market risk becomes individualized. Generous unemployment protection, social spending, and redistribution can buffer the material and psychological consequences of exposure. Where such protection is weaker or inequality is higher, exposure may generate greater demand for redistribution because the risk is less insured.

We use perceived vulnerability to trace how exposure may translate into redistribution demand. Exposure may appear first in expectations of job loss, financial strain, and insufficient income. We therefore advance three hypotheses: H_1 expects higher occupational automation risk to predict more support for redistribution; H_2 expects that association to vary across time; and H_3 expects it to vary across countries. We also examine two mechanisms. M_1 asks whether perceived economic vulnerability partly links automation risk to redistribution demand. M_2 asks whether welfare-state and labor-market institutions moderate the association between exposure, vulnerability, and redistribution.

2.4 An Insurance Model of Redistribution Demand

Appendix 8.1 states a minimal insurance model that yields these hypotheses as explicit conditions. An individual with relative income ξ faces displacement with probability p . Displacement is a temporary spell: the share of income retained in the bad state, δ , is derived from the expected unemployment share, the re-entry wage after displacement, and the level and duration of earnings-related benefits. A linear tax τ funds transfers that accrue at rate θ_g when employed and θ_b when displaced, and preferences are CRRA with risk aversion σ . We interpret the preferred tax rate as demand for redistribution: an individual who prefers a higher τ wants a larger tax-and-transfer program, the disposition the ESS redistribution item elicits. That preferred rate

has a closed form,

$$\tau^* = \frac{\xi (1 - \kappa \delta)}{(\xi - \theta_g) + \kappa (\theta_b - \delta \xi)}, \quad \kappa \equiv \left[\frac{(1 - p) (\xi - \theta_g)}{p (\theta_b - \delta \xi)} \right]^{1/\sigma}, \quad (1)$$

where κ is the consumption ratio across states. Six propositions follow. All are proved analytically; the appendix records the exact expressions and marks which sign regions were mapped numerically.

Two propositions bear directly on the hypotheses. Risk raises demand unconditionally: $\partial \tau^* / \partial p > 0$ for every σ and every interior parameter configuration (Proposition 2). H_1 restates this prediction for occupational automation exposure. The income effect and the income-by-risk interaction are not signed (Propositions 4 and 5). At high risk aversion the income effect becomes positive and insurance behaves as a normal good, the mechanism of Moene and Wallerstein [2001]; at low risk aversion the redistributive motive dominates and the effect is negative, the benchmark of Meltzer and Richard [1981]. The model also nests the closed forms of Alt and Iversen [2017] and Pahontu [2022]. The positive income-by-risk interaction hypothesized by Thewissen and Rueda [2019] requires risk aversion near two or above, together with income-proportional losses or narrow transfer targeting. Meta-analytic estimates of relative risk aversion in consumption contexts center on one after correction for publication bias [Elminejad et al., 2022], so we read that interaction as a joint restriction on primitives rather than an implication of insurance logic.

The remaining propositions organize the mechanism analyses. The model's risk primitive is a perceived displacement probability, so exposure should appear as perceived vulnerability before it appears as policy demand; M_1 examines this link. Propositions 3 and 6 locate the institutional moderation that M_2 examines. The risk slope is steepest where baseline displacement risk is lowest, so institutions that lower that risk steepen the slope through curvature. Flat bad-state transfers weaken the slope. Income retained through earnings-related benefits or rapid re-employment strengthens it. The dividing line is income proportionality, not generosity. H_2 and H_3 follow from the same comparative statics, because these parameters differ across welfare states and change over decades.

2.5 Measuring Routine-Task Automation Risk

The measures used in this literature rate occupations by the routineness of their task content, because computers and machines can more readily perform codifiable, repetitive tasks [Autor and Salomons, 2018]. We use the engineering-bottleneck probability of Frey and Osborne [2017] and the routine-task-intensity (RTI) index of Goos et al. [2014]; they correlate at 0.573 in our sample. Both measures are highest in clerical, production, and lower-skill service occupations, where pay and employment security also tend to be lower.

This is the construct used in the redistribution literature, and we retain it throughout the paper. Frey–Osborne is the reference measure; RTI is the robustness measure. If perceived vulnerability tracks labor-market precarity, routine-task exposure should predict greater vulnerability among respondents in the occupations it identifies as exposed. The automation–redistribution association should be strongest where institutions offer the least protection against that vulnerability.

3 Research Design

3.1 Data and Measures

The analysis uses a harmonized panel of eleven European Social Survey (ESS) waves from 2002 to 2022, spanning 39 countries and roughly 275,000 respondents before listwise deletion. The outcome is the standard ESS item asking whether government should reduce income differences. It is coded on a five-point scale, with higher values indicating greater support for redistribution. Following Thewissen and Rueda [2019], the main analysis is limited to employed and unemployed members of the active labor force.

Figure 1: Distribution of Job Automation Risks in the ESS Sample

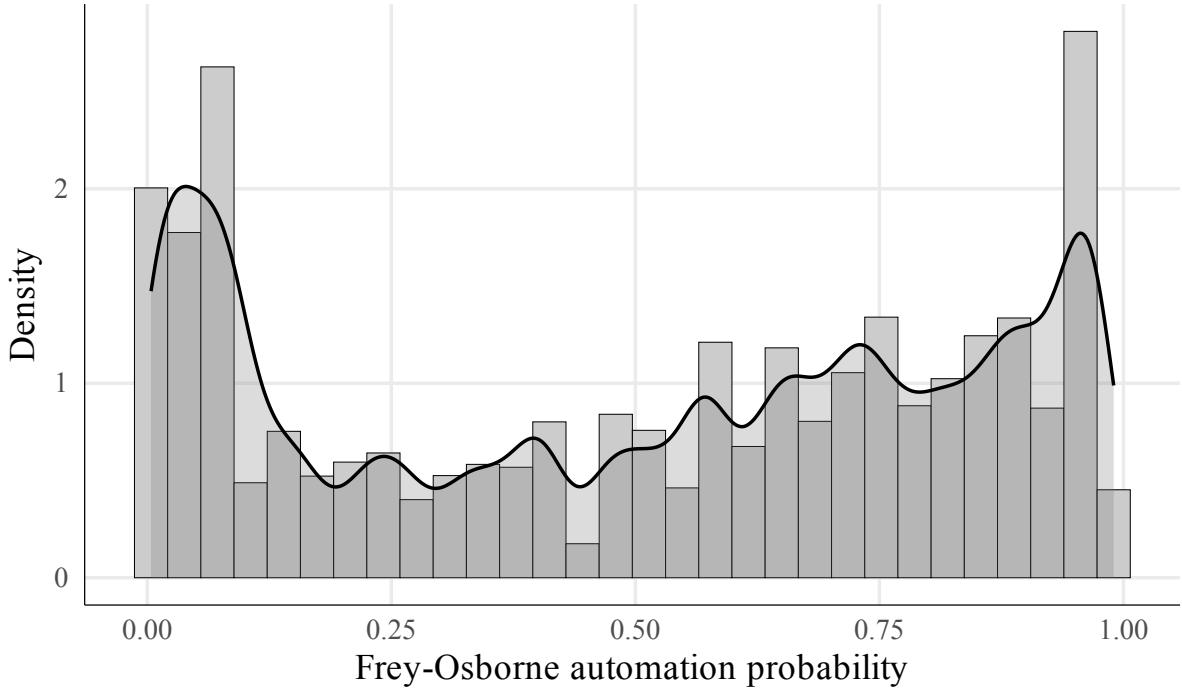


Figure 1 shows Frey–Osborne automation probabilities for ESS occupations.¹ Because this measure is a probability, observations cluster toward the ends of the 0–1 scale. Its density resembles the corresponding U.S. distribution in Frey and Osborne [2017], which is consistent with a successful occupation match.

Frey–Osborne is our reference index, and alongside it we use the Goos–Manning–Salomons routine-task-intensity (RTI) index as a second, independently constructed measure of the same routine-task construct. The panel spans the change in ESS occupation coding: rounds 1–5 use ISCO-88 and rounds 6–11 use ISCO-08, so we harmonize both exposure measures across the two occupation systems.

We measure economic position with income that is equivalized for household size and expressed relative to the country-year mean. We convert income-band responses in ESS show cards to monetary midpoints. For the open top band, we estimate a Pareto mean from the observed counts in the top two categories, following Hout [2004]. We then apply the square-root equivalence scale and divide the resulting value by the design-weighted country-year mean. This control matches the baseline specification in the benchmark literature.

¹The match covers almost 90% of potential cases; matched and unmatched cases do not differ systematically.

We validated the harmonization by joining the raw panel to the original replication data of Thewissen and Rueda [2019] at the respondent level for rounds 1–6. RTI matches exactly, Frey–Osborne probabilities match almost exactly, and the redistribution outcome and core demographic variables have near-perfect agreement. These checks support treating the 2002–2022 panel as a direct extension of the published benchmark design.

The mechanism analyses use several indicators of economic vulnerability. Financial strain is available in every wave and is coded as whether the household finds it difficult or very difficult to cope on its current income. Selected waves provide additional outcomes: expected job loss (a binary indicator for expecting to lose one’s job in the next twelve months) and expected income insufficiency (a binary indicator for expecting income too low to meet needs) in the 2008 and 2016 welfare-attitudes modules, and perceived job insecurity (a scaled measure of the perceived likelihood of job loss) in the 2004 and 2010 work modules. All are restricted to labor-market participants.

At the country level, we examine ten institutional moderators. The Comparative Political Data Set supplies six annual series [CPDS; Armingeon et al., 2025]: union density and collective-bargaining coverage (interpolated from the OECD/AIAS ICTWSS series), employment-protection strictness, left cabinet share, industry employment share, and trade openness. Four additional series come from their source APIs: public social spending as a share of GDP (OECD SOCX), the replacement rate for net unemployment benefits (OECD tax-benefit models; a single person without children at 67% of the average wage, six months into unemployment, without social-assistance top-ups), net income inequality, and relative redistribution (SWIID, version 9.92). The two-stage diagnostic uses their 2016 values, contemporaneous with the country slopes they are correlated with. The moderation models use country-year values that are standardized and lagged by one year. Depending on availability, those models contain 20–39 national clusters.

3.2 Empirical Strategy

For redistribution preferences, we estimate one multilevel model with random effects for each exposure index. It regresses individual support for redistribution on occupational automation exposure and the benchmark controls. The automation-risk slope has global, country, year, and

country-year components (equations 2–3). Every reported redistribution result is derived from that fit. The country and year profiles are random-effect components; the country-year map is the composite $\beta_{c,t}$; and the replication, composition, and institutional diagnostics aggregate or correlate those fitted slopes. We do not re-estimate redistribution coefficients on subsamples.

To ensure consistency with the published benchmark, our primary models include the baseline control vector established by Thewissen and Rueda [2019]. This vector controls for relative equivalized income, years of education, gender, age, current or former union membership, religiosity, and individual unemployment status, adjusted using ESS design weights and accounting for the country-year structure of the survey. The appendix reports sensitivity checks with household size, partner status, and self-placement on the left–right scale.

We estimate the mechanism analyses separately from the redistribution models. At the individual level, mediation models treat financial strain and job insecurity as intervening variables between occupational exposure and support for redistribution. At the country level, each institutional measure is interacted separately with individual automation risk. Models using multiple waves include one-year-lagged moderators, year fixed effects, exposure slopes that vary by country, and a country-year random intercept. The intercept absorbs wave-level macro shocks that would otherwise understate the standard errors of interactions across levels in pooled cross-sections [Schmidt-Catran and Fairbrother, 2016]. These models assess whether protective institutions weaken the association between occupational risk and perceived vulnerability.

Let y_{ict} denote the outcome for individual i in country c in survey year t , and let AR_{ict} denote the occupational automation-risk index. Depending on the analysis, y_{ict} is support for redistribution (the primary outcome), financial strain, expected job loss, expected income insufficiency, or perceived job insecurity. The analytic specification we use for all models is:

$$y_{ict} = \alpha_{ct} + \beta_{ct}AR_{ict} + \mathbf{X}_{ict}\boldsymbol{\phi} + \varepsilon_{ict}, \quad (2)$$

where α_{ct} is a country-year-specific intercept, $\mathbf{X}_{ict}$ is the vector of individual-level controls with coefficient vector $\boldsymbol{\phi}$, and β_{ct} is the country-year-specific slope for automation risk. These slopes are modeled as:

$$\beta_{ct} = \gamma_{10} + u_{1c} + v_{1t} + \omega_{ct}, \quad (3)$$

where γ_{10} is the global baseline slope, u_{1c} and v_{1t} are country and year deviations, and ω_{ct} is the country-year interaction. ε_{ict} is individual-level error clustered by country-year. We estimate this model once per exposure index on the full labor-force sample from 2002 through 2022, using design weights and standardizing exposure on the pooled estimation sample.

The other analyses are instances of the same specification. The vulnerability models replace the outcome with each perceived vulnerability indicator and estimate equation 2 as design-weighted linear probability models with random intercepts and country-clustered standard errors. The institutional moderation models take the Mundlak form:

$$y_{ict} = \alpha_{ct} + \beta_c AR_{ict} + \gamma_B (AR_{ict} \times \bar{Z}_c) + \gamma_W (AR_{ict} \times (Z_{c,t-1} - \bar{Z}_c)) + \mathbf{X}_{ict}\boldsymbol{\phi} + \lambda_t + \varepsilon_{ict}, \quad (4)$$

where $Z_{c,t-1}$ is the lagged institutional moderator, $\bar{Z}_c$ is its country mean, and the within-country variation comes from the deviation $(Z_{c,t-1} - \bar{Z}_c)$. The coefficient γ_B measures between-country moderation and γ_W measures within-country moderation. Main effects of both moderator components, exposure interactions with contemporaneous unemployment and real GDP growth, year fixed effects λ_t , exposure slopes that vary by country, and a country-year random intercept complete the specification.

4 Attenuation and Cross-Country Heterogeneity in the Automation–Redistribution Association

The harmonized ESS panel reproduces the Frey–Osborne association with redistribution in the original 2002–2012 window. We use the benchmark outcome, labor-force restriction, relative-income control, and exposure indices. When the panel is extended through 2022, the aggregate association is smaller. The RTI results are reported in the appendix.

Table 1 summarizes the fitted Frey–Osborne slope by time window. To compare it with the published coefficient, we rescale each estimate to the probability’s 0–1 range. In 2002–2012, the aggregate is 0.175 (95% CI [0.098, 0.251]), which brackets the published benchmark of 0.162. In 2014–2022 it is 0.040 and not distinguishable from zero; the full-panel estimate lies between those values.

The benchmark-roster rows repeat the calculation for the seventeen countries in the published sample. They show the same pattern: the 2002–2012 aggregate is 0.147 (CI [0.069, 0.226]), while the 2014–2022 aggregate is 0.022 and again indistinguishable from zero. Thus, neither replication nor attenuation depends on the broader country coverage of the panel. The final row gives the standardized per-SD aggregate used in the figures. These estimates indicate historical attenuation alongside national variation rather than a uniform disappearance of the relationship.

Table 1: Global FO automation–redistribution association: published benchmark and post-estimation window aggregates of the single heterogeneity model

Quantity	Years	Scale	Estimate	95% CI	N	Countries
Published benchmark	2002–2012	native	0.162	–	64,639	17
Window aggregate of $\beta_{c,t}$	2002–2012	native	0.175	[0.098, 0.251]	104,889	36
Window aggregate of $\beta_{c,t}$	2014–2022	native	0.040	[-0.037, 0.117]	91,749	34
Full-panel aggregate of $\beta_{c,t}$	2002–2022	native	0.112	[0.038, 0.186]	196,638	39
Benchmark-roster aggregate	2002–2012	native	0.147	[0.069, 0.226]	69,788	17
Benchmark-roster aggregate	2014–2022	native	0.022	[-0.057, 0.102]	56,378	16
Benchmark-roster aggregate	2002–2022	native	0.091	[0.016, 0.167]	126,166	17
Full-panel aggregate of $\beta_{c,t}$	2002–2022	per SD	0.038	[0.013, 0.063]	196,638	39

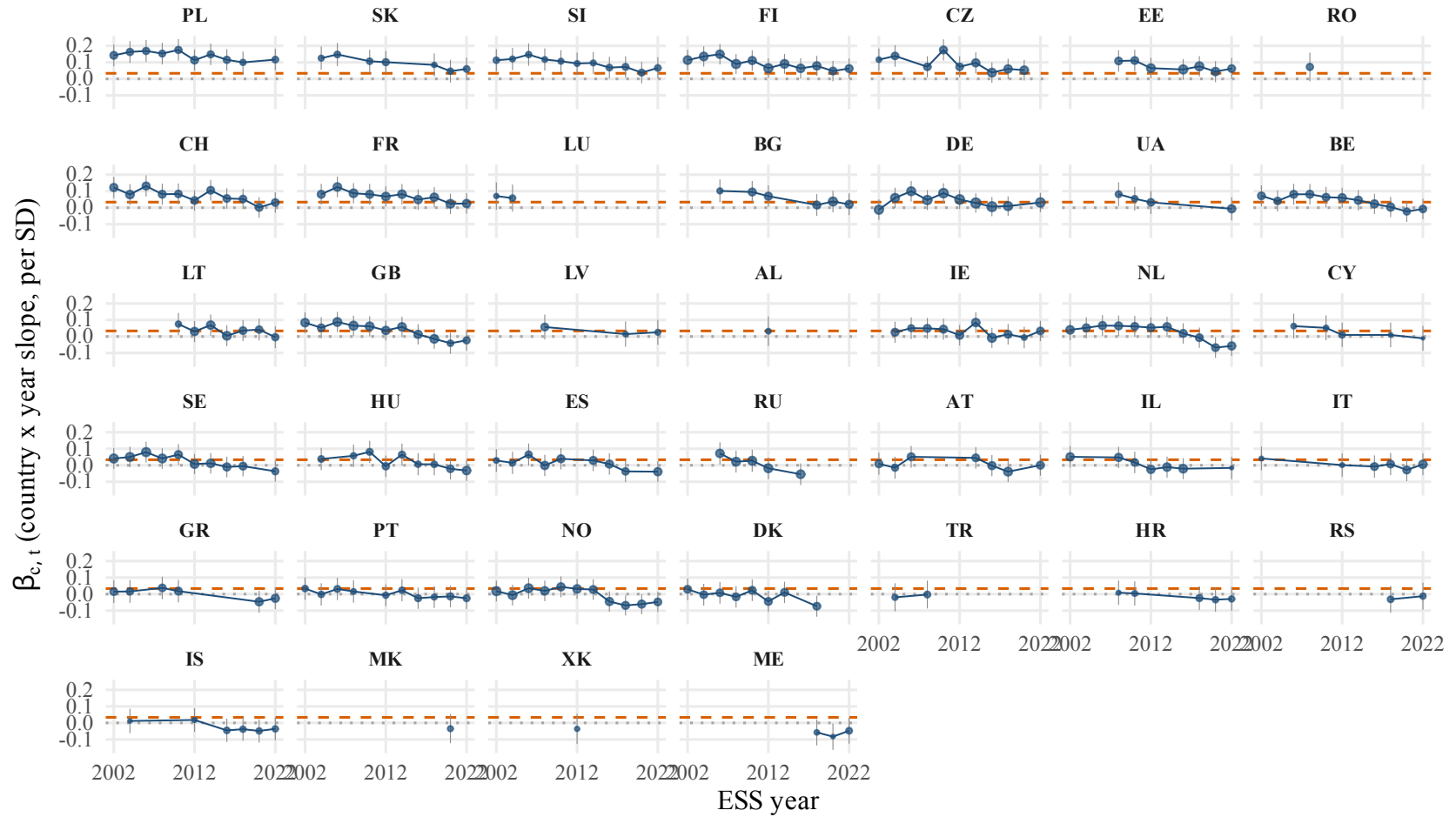
Window aggregates are equal-weighted means of the country-year slopes $\beta_{c,t}$ from the single multilevel heterogeneity model; no additional model is estimated. The model adjusts for relative equivalized household income, years of education, gender, age, union membership, religiosity, and unemployment status, with ESS design weights. Benchmark-roster rows restrict the aggregation to the seventeen countries of the published benchmark’s sample, holding country composition fixed at the benchmark roster. Native-scale entries divide the per-SD aggregate by the pooled exposure SD. Intervals carry over the fixed-slope sampling variance and the aggregated conditional variances of the slope components; cross-component posterior covariances are ignored. The published-benchmark row is quoted from Thewissen and Rueda (2019) as an external reference.

Figure 2 displays estimates for every country-year. They range from negative to strongly positive values. Country-level variation exceeds temporal variation, so the association is more stable within countries than across survey waves. This pattern is consistent with the proposition that national settings condition how occupational exposure relates to redistribution preferences; it does not by itself establish an institutional cause. The appendix provides the corresponding RTI plot.

Figure 2: Country-by-year heterogeneity in the automation–redistribution association

Frey-Osborne (FO) slope by country over time

$\beta_{c,t} = \gamma + u_c + v_t + w_{ct}$ from the single model; panels ordered by mean slope; point size $\sim$ cell N



Dashed orange = global slope (0.034); dotted = 0. Whiskers: 95% intervals combining the fixed-slope sampling variance with the conditional variances of the country, year, and country-year slope components.

Figure 3 separates the temporal pattern from changing country coverage. It aggregates the same fitted $\beta_{c,t}$ by year for four country rosters: all observed countries, a common panel of countries present in at least three waves with a pre-2016 anchor, the seventeen countries in the Thewissen and Rueda [2019] sample, and a balanced panel of the eleven countries present in every wave. Equal country weights make differences among curves attributable to roster composition; no coefficient is re-estimated.

The four series differ in level, with the balanced panel showing the largest slopes, but each declines after 2014, reaches a low point in the COVID-disrupted 2020 round, and partly recovers in 2022. The pattern therefore does not appear to be driven by changing country coverage. The 95% intervals combine the fixed-slope and year-component variance with averaged conditional variances for country and country-year components. The appendix reports the RTI counterpart.

Figure 3: Yearly Frey–Osborne automation–redistribution slopes under four composition assumptions

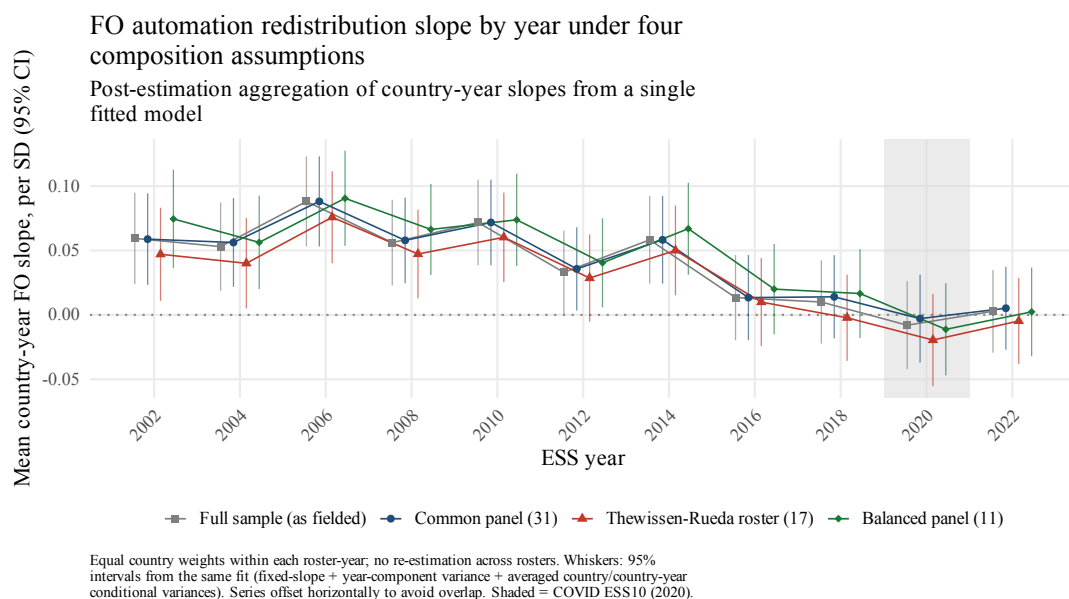


Figure 4 reports the marginal effect of Frey–Osborne exposure in the latest available observation for each country. The figure labels the year because nine of the 39 countries were last observed before 2022. Most associations are not statistically distinguishable from zero. Poland (0.117 per standard deviation), Slovenia (0.066), and Finland (0.062) have positive significant effects, while Denmark (−0.073) has a negative significant effect. The range, from −0.07 to 0.12 per standard deviation, shows that the post-2014 association is specific to particular national

contexts; there is no general European pattern. The appendix reports the RTI counterpart.

Figure 4: Country marginal effects of Frey–Osborne exposure at the most recent available wave

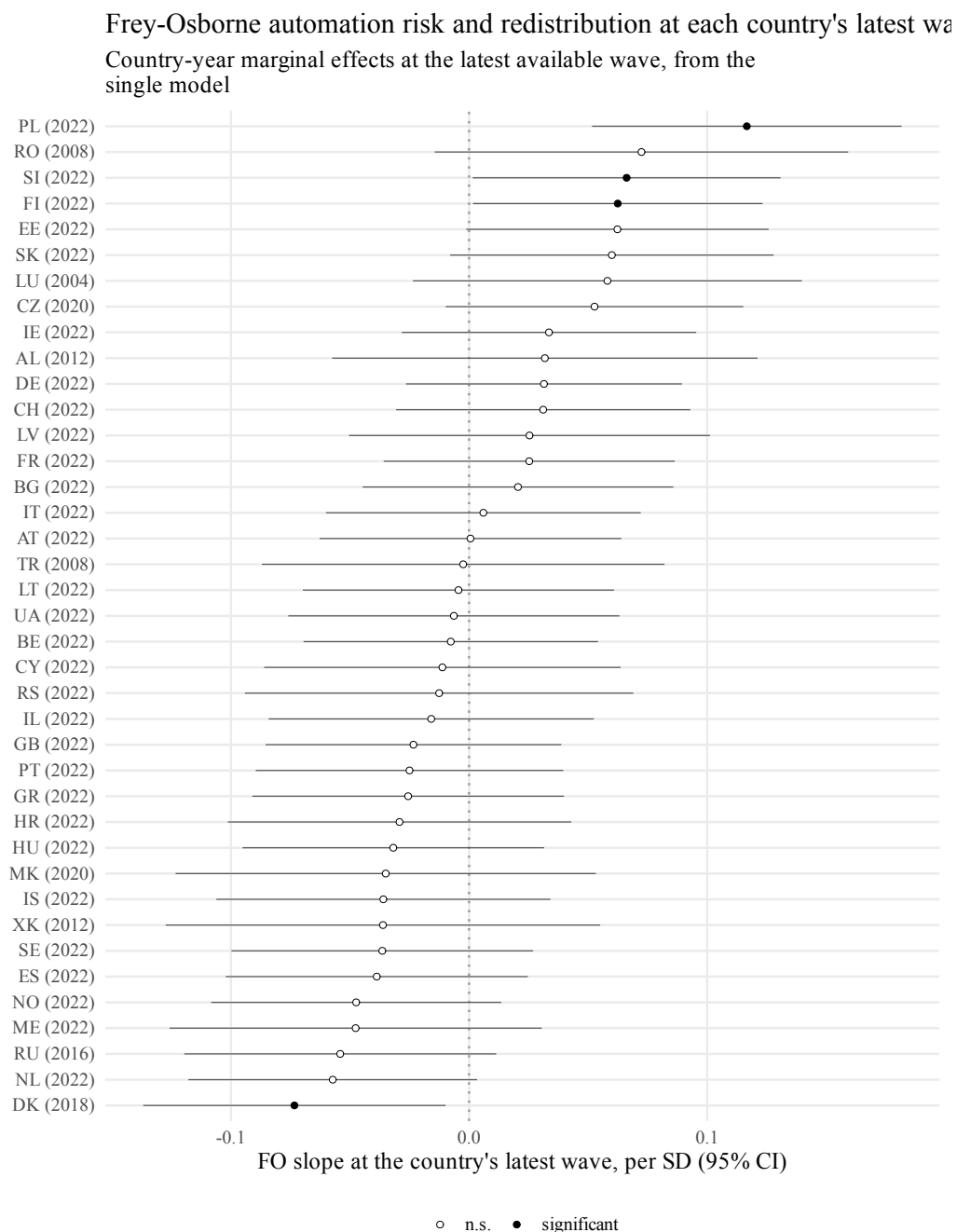
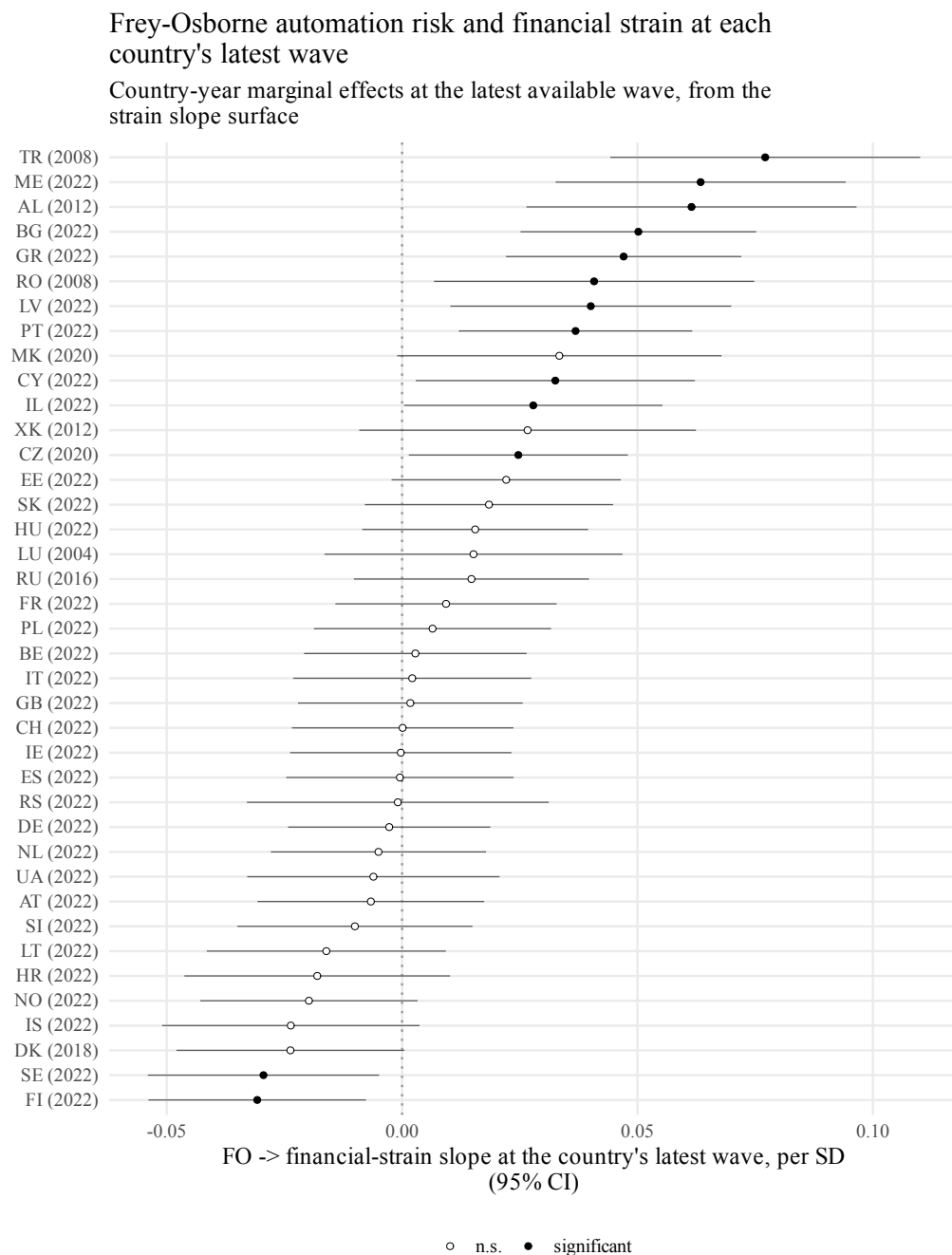


Figure 5 repeats the exercise with financial strain as the outcome. The two margins diverge. At the latest wave, the exposure–strain slope is positive and significant in ten countries, concentrated in Southern and Eastern Europe, while the exposure–redistribution slope is positive in three. The countries also differ: Portugal, Greece, Bulgaria, and Latvia show strain without

demand, and Finland shows demand without strain; in Finland and Sweden the strain slope is negative. Under the model, the first pattern is consistent with uninsured exposure that still binds materially where demand has faded, and the second with institutions that absorb the material consequence while political demand persists. Attribution country by country is beyond the design.

Figure 5: Country marginal effects of Frey–Osborne exposure on financial strain at the most recent available wave



5 Mechanism Analyses: Vulnerability and Institutional Moderation

The paper's main estimand is the association between automation exposure and redistribution preferences. The analyses in this section examine mechanisms behind that association. At the individual level, we ask whether automation exposure is associated with perceived economic vulnerability. At the country level, we ask whether welfare-state and labor-market institutions moderate the exposure–vulnerability link, helping explain why the redistribution association is stronger in some national contexts than others.

Figure 6 presents the individual-level mechanism results. Across all eleven waves, automation exposure is positively associated with financial strain. In the rounds that include them, exposure is also positively associated with expected job loss, expected income insufficiency, and job insecurity. Appendix Table 5 reports the corresponding estimates for the 2008 and 2016 welfare-module rounds; the 2016 columns are the confirmatory specification and the 2008 columns are an extension under the same model. These outcomes help assess whether occupational exposure is linked to the perceived vulnerability through which it may shape redistribution preferences.

Figure 6: Automation exposure and perceived economic vulnerability across ESS waves

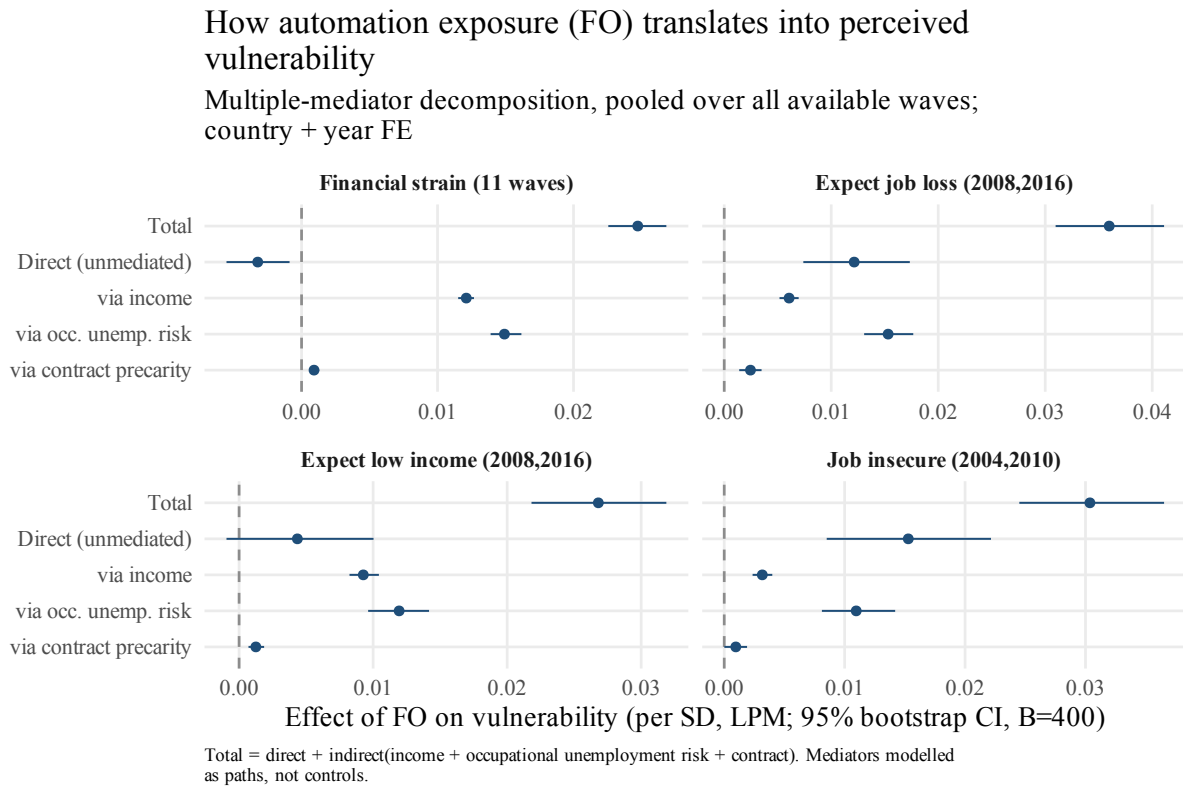


Figure 7 examines whether institutions condition the association between automation risk and financial strain. The models use one-year-lagged moderators, year fixed effects, a country-year random intercept, and business-cycle adjustment. Survey-year unemployment and real GDP growth enter as both main effects and interactions with exposure.

This adjustment is needed because benefits and spending expand in downturns, when the exposure–strain gradient also steepens. Without adjustment, this countercyclical co-movement obscures a buffering pattern. Net of the cycle, within-country estimates resemble the between-country diagnostic (two-stage correlations of -0.4 to -0.7 for union density, bargaining coverage, social spending, and realized redistribution; $+0.67$ for net inequality): union density weakens the exposure–strain association for both exposure measures; bargaining coverage and social spending show the same direction for RTI; and net income inequality and employment-protection strictness strengthen the association for Frey–Osborne exposure. The last result is consistent with insider–outsider dualization, in which strict protection shelters insiders while leaving exposed outsiders more precarious. Because ten moderators are tested separately with roughly thirty national clusters, these results are evidence consistent with institutional condi-

tioning; the design does not yield definitive causal estimates.

Figure 7: Institutional moderation of the automation–financial-strain association

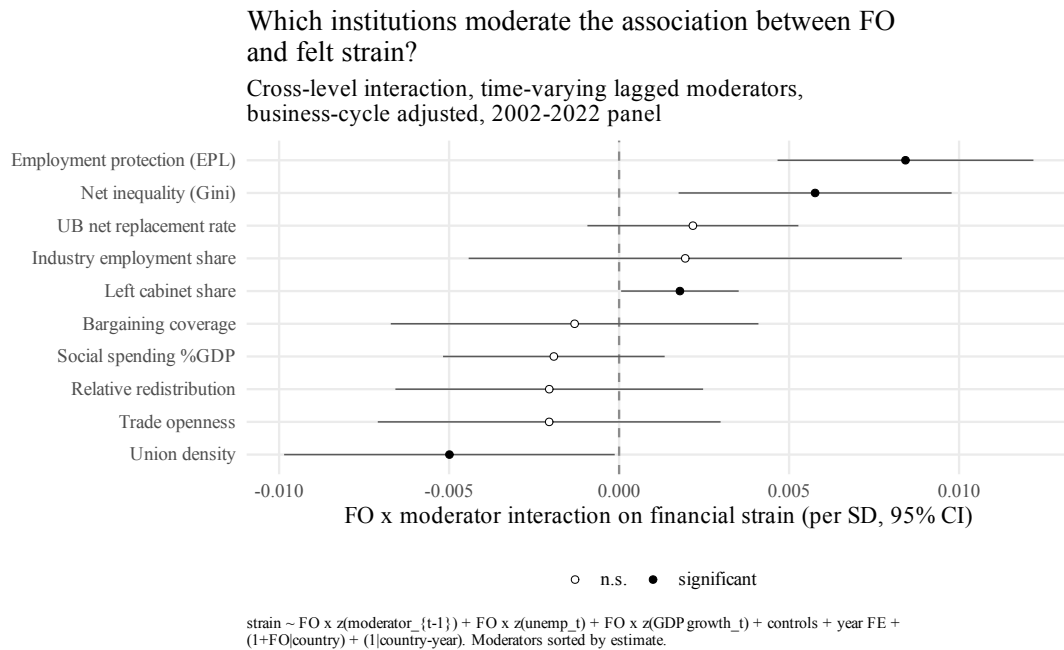


Table 2 separates each institutional interaction with the redistribution gradient into a between-country and a within-country component. The Mundlak specification (equation 4) enters the lagged moderator as its country mean and as the deviation from that mean, and both terms interact with exposure in the same model. Four results replicate across both exposure indices at a Benjamini–Hochberg threshold of 0.10. The between-country interaction for net inequality is negative, the within-country interactions for union density and bargaining coverage are positive, and the within-country interaction for trade openness is negative. The within-country components are the stronger evidence, because they are identified from changes relative to a country’s own average, so time-invariant country confounds drop out; a within-country coefficient explains the temporal attenuation only if the moderator itself also moved.

Table 2: Between- and within-country institutional moderation of the Frey–Osborne automation–redistribution gradient (Mundlak decomposition)

Moderator	Between country		Within country		N	Countries
	Est.	(SE)	Est.	(SE)		
Net inequality (Gini)	-0.0192*	(0.0075)	-0.0066*	(0.0026)	194,402	38
Social spending %GDP	+0.0006	(0.0083)	-0.0131***	(0.0024)	183,477	31
Relative redistribution	+0.0149	(0.0079)	+0.0013	(0.0026)	189,303	33
UB net replacement rate	-0.0092	(0.0074)	+0.0052	(0.0027)	179,834	31
Left cabinet share	-0.0050	(0.0077)	+0.0078**	(0.0023)	179,126	30
Union density	-0.0150	(0.0075)	+0.0156***	(0.0032)	150,814	30
Bargaining coverage	-0.0137	(0.0076)	+0.0184*	(0.0067)	150,560	30
Employment protection (EPL)	+0.0037	(0.0075)	-0.0007	(0.0030)	146,207	25
Industry employment share	+0.0050	(0.0075)	+0.0091**	(0.0026)	179,126	30
Trade openness	+0.0040	(0.0080)	-0.0116**	(0.0032)	160,954	30
Benefit earnings-relatedness	+0.0170*	(0.0069)	-0.0048	(0.0025)	179,834	31

Each row is one multilevel model of support for redistribution on standardized Frey–Osborne exposure, following the Mundlak (hybrid) specification: the moderator, lagged one survey year, is split into its country mean (between component) and the deviation from that mean (within component), both standardized, and both interacted with exposure in the same model. Cells report the cross-level interaction coefficients. All models include exposure interactions with contemporaneous unemployment and real GDP growth, the baseline individual controls, year fixed effects, a random exposure slope by country, and a country-year random intercept, with ESS design weights and maximum likelihood estimation. Stars mark Benjamini–Hochberg adjusted p values, corrected across the full family of 44 coefficients (eleven moderators, two components, two exposure indices): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Additional mechanism diagnostics, robustness checks, and full auxiliary tables appear in Appendix 8.2.

6 Discussion

The findings qualify a simple, time-invariant account of the political consequences of technological change, because the association between automation risk and redistribution preferences reproduces in the pre-2016 period yet attenuates in the subsequent decade.

Individual countries illustrate the pattern (Figure 2). France shows significant positive slopes in seven of eight rounds from 2004 through 2018 and null estimates in 2020 and 2022. The same fade arrives earlier in Great Britain: significant positive slopes through 2010 become null estimates from 2012 onward, and the point estimates run negative after 2016. Finland is the

counterexample, with a significant positive slope in nearly every round including 2022. Spain never shows a significant slope in any round. Under the model, such divergence is what heterogeneous insurance parameters produce, although attribution of any single country's path to a specific institution is beyond the design.

One possible interpretation is a shift in the distribution of technological risk. Routine-biased technological change in the 2000s was concentrated among clerical and manual occupations, whereas machine learning and AI may expose professionals with higher skills and incomes to technological precarity [Gallego and Kurer, 2022]. In fact, Appendix Figure 13 documents the measured part of this shift: design-weighted routine exposure declined across rounds for the benchmark roster, and exposure specific to newer AI lies outside both indices. Hence professionals newly exposed to this risk may have weaker material incentives for conventional income redistribution, and they may instead favor instruments such as social investment or basic income [Busemeyer and Sahm, 2022].

The mechanism analyses indicate why the redistribution association may concentrate in particular national contexts. To begin with, automation risk is associated with perceived vulnerability, including financial strain and expected job loss. Whether institutions buffer that vulnerability is harder to see, because welfare effort expands countercyclically, in the same downturns that steepen the exposure–strain association. Unadjusted interaction estimates can therefore take signs opposite to buffering, and the institutional pattern becomes clear only once business-cycle dynamics enter the models.

After accounting for unemployment and growth, the results are consistent with the power-resources account [Stephens, 1979] and with the social-insurance account [Moene and Wallerstein, 2001]. For example, union density weakens the exposure–strain association within and between countries, and bargaining coverage and social spending have the same direction for RTI. In contrast, net income inequality and strict employment protection strengthen the Frey–Osborne association. Yet these estimates are observational, test one moderator at a time, and use roughly thirty national clusters. Hence they should be read as institutional patterns consistent with theory; the design does not identify causal effects.

We interpret these estimates through the model's comparative statics. Its unconditional pre-

diction is that displacement risk raises the preferred tax rate, and the replication recovers that association. It does not sign the income effect or the income-by-risk interaction at empirically plausible risk aversion, so we treat the benchmark interaction as a joint restriction on primitives. Moreover, the direction of the social-spending result matches the flat-transfer proposition. The curvature proposition suggests an interpretation of the employment-protection result in which institutions that lower baseline displacement risk steepen the risk slope, although the displacement-probability channel is not measured here. However, two primitives behind the bad-state income share, the re-entry wage after displacement and the expected duration of unemployment, remain untested in this design, and we identify them as priorities for future measurement.

Finally, several limitations of this study warrant caution. First, our operationalization of automation risk relies on occupational-level indices (Frey–Osborne probabilities and RTI), which assume task structures are uniform within broad occupations. Second, the harmonization of occupations across the ISCO-88 and ISCO-08 systems introduced potential crosswalk measurement error. Third, changing country coverage across different ESS waves prevents us from estimating a perfectly balanced panel, though we address this via composition-adjusted common-sample models. Fourth, our observational design limits our ability to make causal claims, and we treat the associations as descriptive patterns that vary with institutional context.

7 Conclusion

Using harmonized European Social Survey data for 2002–2022, this paper reexamined the relationship between occupational automation risk and demand for redistribution. The replication returns three findings. First, the positive association reappears in the 2002–2012 window and brackets the published benchmark. Second, the association weakens after 2014 under every roster of countries in the analysis. Third, variation across countries exceeds variation over time, so national settings, more than common shocks, condition the relationship.

The mechanism analyses add two regularities. Automation exposure predicts financial strain, expected job loss, and perceived job insecurity in every wave with the relevant items. Welfare-

state and labor-market institutions condition the exposure–strain link once business-cycle dynamics enter the models: union density weakens it within and between countries, while net income inequality and strict employment protection strengthen it, a pattern consistent with insider–outsider dualization.

A formal insurance model gives these findings a common structure. Displacement risk raises the preferred tax rate at every interior parameter configuration, and the income effect and its interaction with risk depend on risk aversion. Institutional moderation depends on the composition of income in the bad state. Under this interpretation, the attenuation and the cross-country heterogeneity reflect parameters that vary across welfare states and change slowly over decades.

Three directions for future research follow from these results. First, the routine-task indices date from the era of routine-biased computerization; exposure measures built for generative AI would show whether the attenuation reflects the measure or the politics. Second, the model marks the re-entry wage after displacement and the expected duration of unemployment as the unmeasured primitives behind institutional moderation, and both belong in future surveys. Third, designs built around institutional reforms would improve identification over comparisons across countries.

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8 Appendix

8.1 A Formal Insurance Model of Redistribution Preferences

This appendix states the model behind Section 2.4 and gives the exact expressions for its comparative statics. The propositions are proved analytically and verified symbolically; the remark at the end records which sign regions were mapped numerically.

Primitives. An individual has pre-tax income y ; mean income is $\bar{y}$, and relative income is $\xi \equiv y/\bar{y}$. With probability p she is displaced; with probability $1 - p$ she keeps her job and earns y . Displacement is a temporary spell. Let $\mu \in [0, 1]$ be the expected share of the period spent unemployed, $\rho \in [0, 1]$ the re-entry wage ratio (so $1 - \rho$ is scarring), b the earnings-related benefit replacement rate, and B the benefit duration as a share of the period. Income in the bad state is δy , with

$$\delta = (1 - \mu)\rho + b \cdot \min(\mu, B). \quad (5)$$

δ is a derived quantity with four separately observable components; it is not a free parameter summarizing protection. Interpretation depends on the distinction: each institution enters through the component on which it acts.

Policy. A linear tax $\tau \in [0, 1]$ on income funds transfers that accrue at rate θ_g in the good state and θ_b in the bad state, with $\theta_b \geq \theta_g \geq 0$. Income proportionality separates δ from θ : δ collects payments that scale with the individual's own prior earnings and are exogenous to her policy choice, while θ collects flat payments that scale with the chosen tax rate. Earnings-related unemployment insurance therefore belongs in δ ; flat-rate benefits and social assistance belong in θ_b . Consumption is affine in τ :

$$c_g = \xi - \tau(\xi - \theta_g), \quad c_b = \delta\xi + \tau(\theta_b - \delta\xi). \quad (6)$$

Preferences and objective. Utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$ for $\sigma > 0$, with $u(c) = \ln c$ at $\sigma = 1$. The individual maximizes

$$V(\tau) = (1-p)u(c_g) + pu(c_b). \quad (7)$$

CRRA is homothetic, so only relative income matters; we set $\bar{y} = 1$. This matches the measurement practice in the empirical sections, where the income control is equivalized income relative to the country-year mean.

Interior region. Assume

$$\delta\xi < \theta_b, \quad \theta_g < \xi. \quad (8)$$

The second inequality makes the individual a net contributor when employed; the first makes her a net recipient when displaced. Together they generate an insurance trade-off; outside this region the problem has a corner solution.

Solution and comparative statics. Write $A \equiv \xi - \theta_g$, $C \equiv \theta_b - \delta\xi$, $G \equiv \theta_b - \delta\theta_g$, and $D \equiv A + \kappa C$.

Proposition 1 (Optimal tax rate). *Under (8), V is strictly concave in τ and the optimum is*

$$\tau^* = \frac{\xi(1-\kappa\delta)}{D}, \quad \kappa \equiv \left[\frac{(1-p)A}{pC} \right]^{1/\sigma}, \quad (9)$$

interior whenever $\kappa\delta < 1$ and $\tau^ < 1$.*

Proof. The first-order condition $V'(\tau) = 0$ gives $(c_g/c_b)^\sigma = (1-p)A/(pC)$, so $c_g = \kappa c_b$. $V''(\tau) = -\sigma(1-p)A^2c_g^{-\sigma-1} - \sigma pC^2c_b^{-\sigma-1} < 0$, so the critical point is the global maximum. Substituting (6) into $c_g = \kappa c_b$ and solving for τ yields (9). $\square$

The auxiliary variable $\kappa = c_g/c_b \geq 1$ is the risk the individual still bears at her chosen policy; each comparative static below depends on it. The risk slope, the model analogue of the estimated

slopes β_{ct} , also has a closed form:

$$\beta \equiv \frac{\partial \tau^*}{\partial p} = \frac{\xi G \kappa}{\sigma p (1 - p) D^2}. \quad (10)$$

Proposition 2 (Risk raises demand unconditionally). *If $\theta_b > \delta \theta_g$, then $\partial \tau^* / \partial p > 0$ throughout the interior region, for every $\sigma > 0$ and $\delta \in [0, 1)$.*

Proof. $\partial \tau^* / \partial \kappa = -\xi G / D^2 < 0$ and $d\kappa / dp = -\kappa / [\sigma p (1 - p)] < 0$; the product is (10), which is positive. $\square$

The condition $\theta_b > \delta \theta_g$ is implied by $\theta_b \geq \theta_g$ and $\delta < 1$, so it does not bind in any case of interest. This is the model's only unconditional prediction, and it requires no restriction on risk aversion.

Proposition 3 (The risk slope declines in baseline risk). *With $r \equiv A / (\kappa C)$,*

$$\text{sign} \left(\frac{\partial \beta}{\partial p} \right) = \text{sign} \left\{ (2p - 1) + \frac{1 - r}{\sigma(1 + r)} \right\}. \quad (11)$$

Sufficient for a negative sign: $p \leq 1/2$ and $r \geq 1$.

At empirically relevant displacement probabilities the slope is steepest where baseline risk is lowest. This is a channel distinct from the composition of the bad state: an institution that lowers the level of displacement risk, employment protection being the leading case, steepens the exposure gradient through curvature without changing what a displaced worker retains.

Proposition 4 (The income effect is not signed).

$$\frac{d\tau^*}{d\xi} = \frac{(\kappa \theta_b - \theta_g)(1 - \kappa \delta) - \xi G \kappa'_\xi}{D^2}, \quad \kappa'_\xi = \frac{\kappa}{\sigma} \left(\frac{1}{A} + \frac{\delta}{C} \right). \quad (12)$$

The two terms oppose each other, and the balance depends on σ . As $\sigma \rightarrow \infty$, κ'_ξ vanishes and the effect is positive: insurance behaves as a normal good, the region of Moene and Wallerstein [2001]. At low σ the second term dominates and the effect is negative, the benchmark of Meltzer and Richard [1981]. Under full targeting ($\theta_g = 0$) with log utility, income drops out of (9) entirely, the knife-edge of Alt and Iversen [2017].

Proposition 5 (The income-by-risk interaction is not generic).

$$\frac{\partial \ln \beta}{\partial \xi} = \frac{1}{\xi} + \frac{\kappa'_\xi}{\kappa} - \frac{2(1 - \kappa\delta + \kappa'_\xi C)}{D}. \quad (13)$$

The sign is a joint restriction on risk aversion, loss severity, and transfer targeting. Positive signs concentrate at σ near two or above with δ bounded away from zero, or under narrow targeting.

Proposition 6 (Transfers and retention move the slope in opposite directions).

$$\frac{\partial \ln \beta}{\partial \theta_b} = \frac{1}{G} - \frac{1}{\sigma C} - \frac{2\kappa(1 - 1/\sigma)}{D}, \quad \frac{\partial \ln \beta}{\partial \delta} = -\frac{\theta_g}{G} + \frac{\xi}{\sigma C} + \frac{2\kappa\xi(1 - 1/\sigma)}{D}. \quad (14)$$

Over the interior grids, raising the flat bad-state transfer θ_b lowers β in 98.9% of draws, and raising the retention share δ raises β in 100%. Holding the induced gain in bad-state consumption identical across the two routes, the flat route buffers and the income-proportional route amplifies in 100% of cases.

The last result is both a prediction and an identification warning: the same unemployment-insurance policy buffers the exposure gradient if written as a transfer and amplifies it if written as retained income. Institutions must therefore be mapped through the primitives in (5), not through policy labels. Unions and bargaining coverage act on several primitives at once (ρ , the separation rate, and θ_b) and are deliberately left unsigned. Benefit duration B is also unsigned once its behavioral effect on μ is allowed to vary over the range documented in quasi-experimental work, and the re-entry wage ratio ρ has no country-year series; of the five primitives, three can be signed and two cannot.

Remark 1. The expressions above are exact and were verified symbolically (sympy); the closed form (9) was checked against direct numerical maximization of (7) on 2,533 interior draws with no discrepancy above 10^{-5} . The sign regions reported for Propositions 3 through 6 were mapped numerically over stated parameter grids.

Nested cases. As $p \rightarrow 0$ the insurance motive vanishes and only the redistributive motive survives [Meltzer and Richard, 1981]. Setting $\theta_g = 0$, $\delta = 0$, $\sigma = 1$ collapses (9) to $\tau^* = p$, the result of Alt and Iversen [2017]: the preferred tax rate equals the probability of falling

into poverty, independent of own income. Setting $\theta_g = \gamma$, $\theta_b = 1 - \gamma$, $\delta = 0$, $\sigma = 1$ gives $\tau^* = p\xi/(\xi - \gamma)$, the closed form of Pahontu [2022]. High σ with $\delta \approx 0$ yields $d\tau^*/d\xi > 0$, the region of Moene and Wallerstein [2001]. Meta-analytic estimates of relative risk aversion in consumption contexts center on one after correction for publication bias [Elminejad et al., 2022]; at that value the Moene–Wallerstein region and the positive income-by-risk interaction both lie outside the admissible parameter space.

Mapping to automation exposure. Let a denote occupational automation exposure. With $p = p(a)$, $\xi = \xi(a)$, $\delta = \delta(a)$,

$$\frac{d\tau^*}{da} = \frac{\partial\tau^*}{\partial p} \frac{dp}{da} + \frac{\partial\tau^*}{\partial\xi} \frac{d\xi}{da} + \frac{\partial\tau^*}{\partial\delta} \frac{d\delta}{da}, \quad (15)$$

so an estimated exposure coefficient is a composite of a risk channel, an expected-wage channel, and a severity channel. The model signs only the risk channel. This is one reason the paper does not treat its mechanism analyses as tests.

8.2 Supplementary Figures and Tables

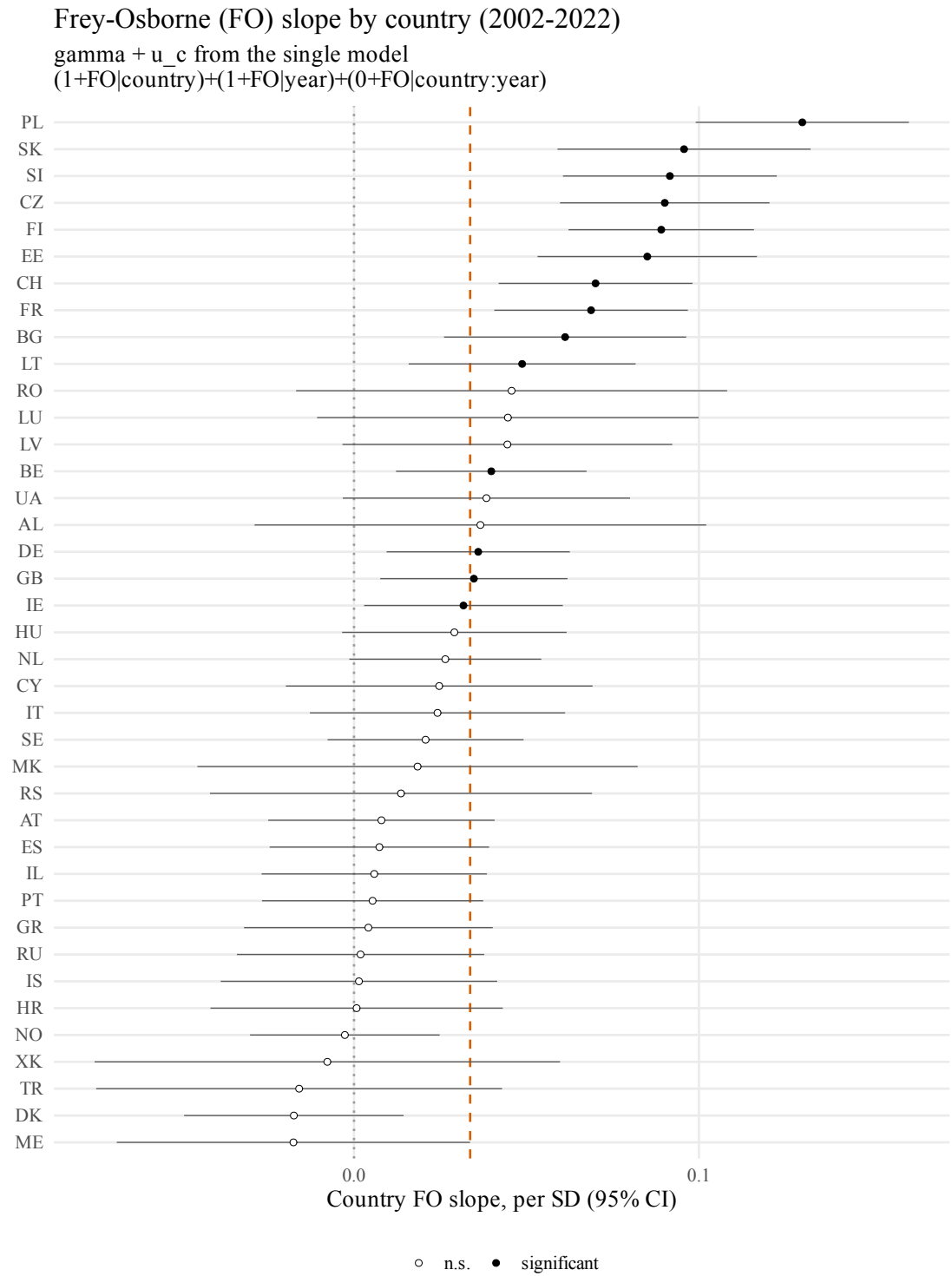
This appendix reports RTI robustness results, country and year slope profiles, composition aggregates, latest-wave country estimates, mediation diagnostics, a single-wave diagnostic of exposure and perceived vulnerability, and the institutional two-stage diagnostic.

Table 3: Global RTI automation–redistribution association: published benchmark and post-estimation window aggregates of the single heterogeneity model

Quantity	Years	Scale	Estimate	95% CI	<i>N</i>	Countries
Published benchmark	2002–2012	per SD	0.042	–	64,639	17
Window aggregate of $\beta_{c,t}$	2002–2012	native	0.039	[0.020, 0.058]	98,068	36
Window aggregate of $\beta_{c,t}$	2014–2022	native	0.015	[-0.004, 0.035]	83,543	34
Full-panel aggregate of $\beta_{c,t}$	2002–2022	native	0.028	[0.010, 0.046]	181,611	39
Benchmark-roster aggregate	2002–2012	native	0.042	[0.022, 0.062]	65,134	17
Benchmark-roster aggregate	2014–2022	native	0.019	[-0.001, 0.039]	51,216	16
Benchmark-roster aggregate	2002–2022	native	0.032	[0.013, 0.050]	116,350	17
Full-panel aggregate of $\beta_{c,t}$	2002–2022	per SD	0.026	[0.009, 0.043]	181,611	39

Window aggregates are equal-weighted means of the country-year slopes $\beta_{c,t}$ from the single multilevel heterogeneity model; no additional model is estimated. The model adjusts for relative equivalized household income, years of education, gender, age, union membership, religiosity, and unemployment status, with ESS design weights. Benchmark-roster rows restrict the aggregation to the seventeen countries of the published benchmark’s sample, holding country composition fixed at the benchmark roster. Native-scale entries divide the per-SD aggregate by the pooled exposure SD. Intervals carry over the fixed-slope sampling variance and the aggregated conditional variances of the slope components; cross-component posterior covariances are ignored. The published-benchmark row is quoted from Thewissen and Rueda (2019) as an external reference.

Figure 8: Marginal country slopes from the single heterogeneity model, Frey–Osborne



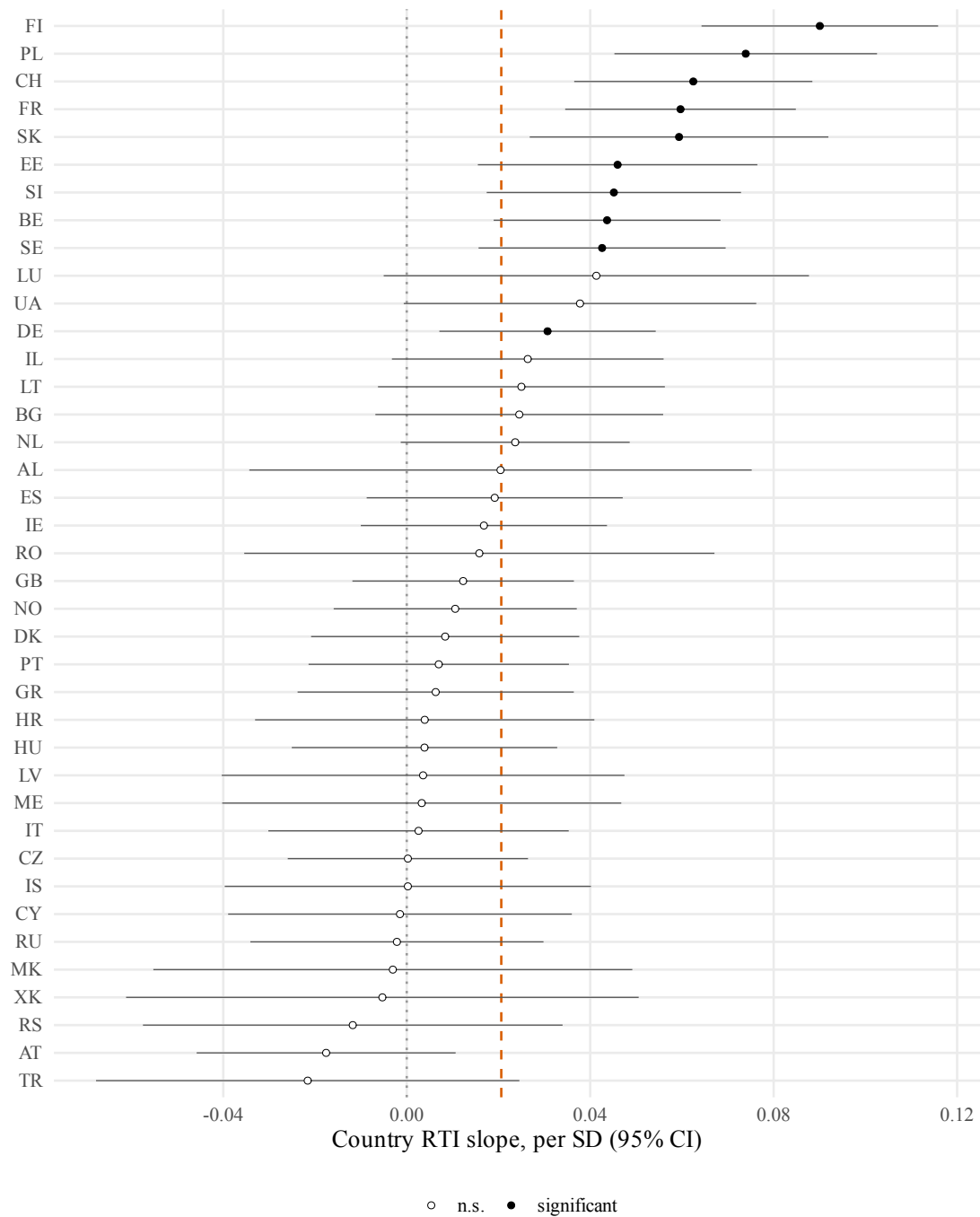
Dashed orange = global slope (0.034); dotted = 0.

Figure 9: Marginal country slopes from the single heterogeneity model, RTI

Routine-task intensity (RTI) slope by country (2002-2022)

gamma + u_c from the single model

(1+RTI|country)+(1+RTI|year)+(0+RTI|country:year)

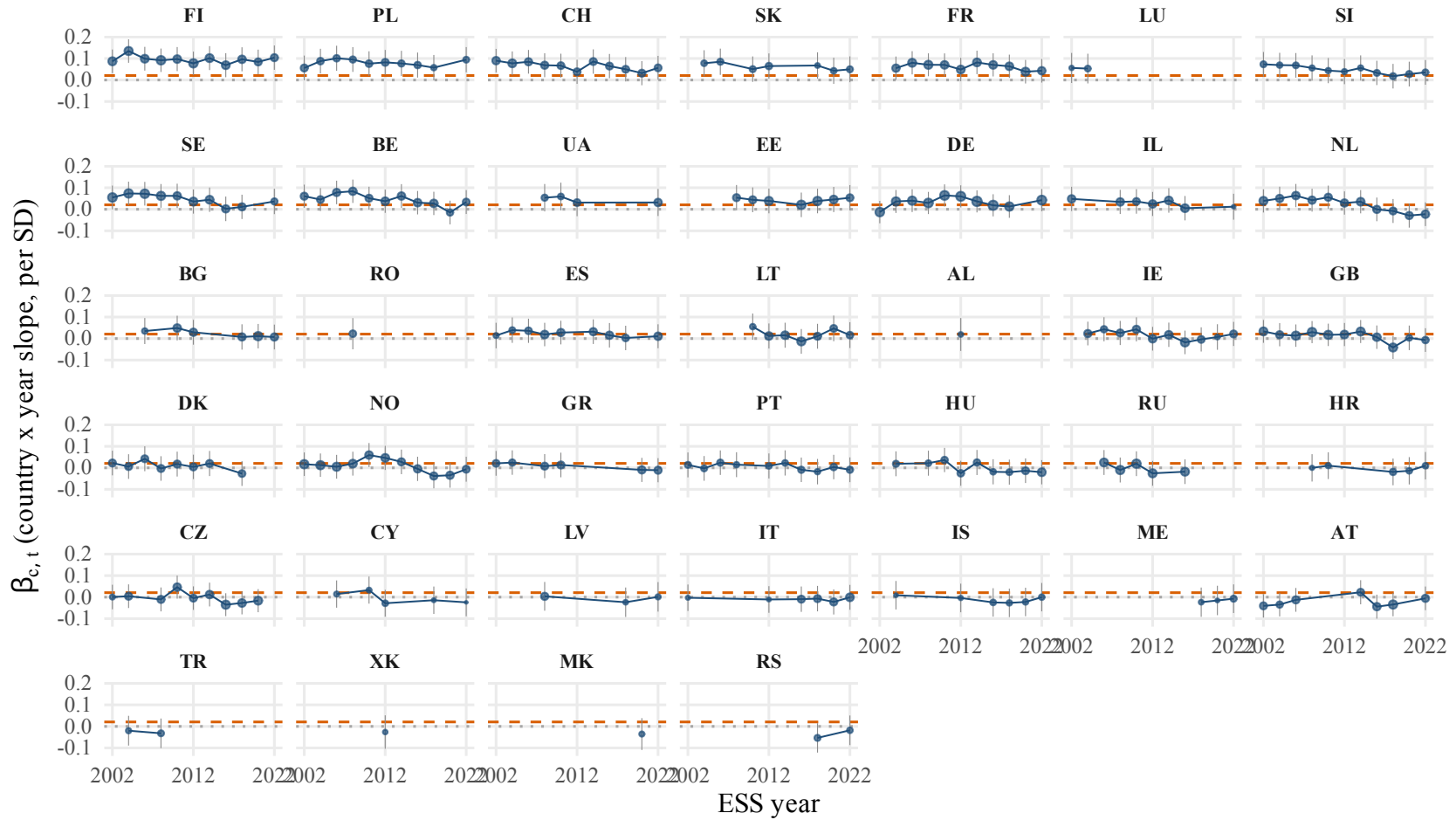


Dashed orange = global slope (0.021); dotted = 0.

Figure 10: RTI robustness: country-by-year heterogeneity in the automation–redistribution association

Routine-task intensity (RTI) slope by country over time

$\beta_{c,t} = \gamma + u_c + v_t + w_{ct}$ from the single model; panels ordered by mean slope; point size $\sim$ cell N



Dashed orange = global slope (0.021); dotted = 0. Whiskers: 95% intervals combining the fixed-slope sampling variance with the conditional variances of the country, year, and country-year slope components.

Figure 11: Marginal year slopes from the single heterogeneity model (composition-confounded)

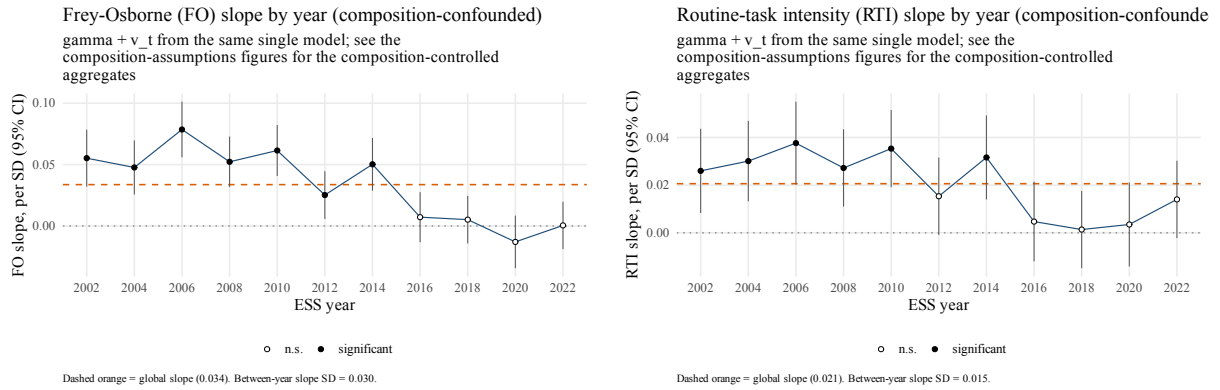


Figure 12 gives the RTI counterpart to Figure 3. It aggregates the fitted RTI country-year slopes $\beta_{c,t}$ over the same four rosters and constructs intervals in the same way. The RTI series also differ in level across rosters, decline after 2014, and partly recover in 2022.

Figure 12: RTI robustness: yearly mean country-year slopes under four composition assumptions

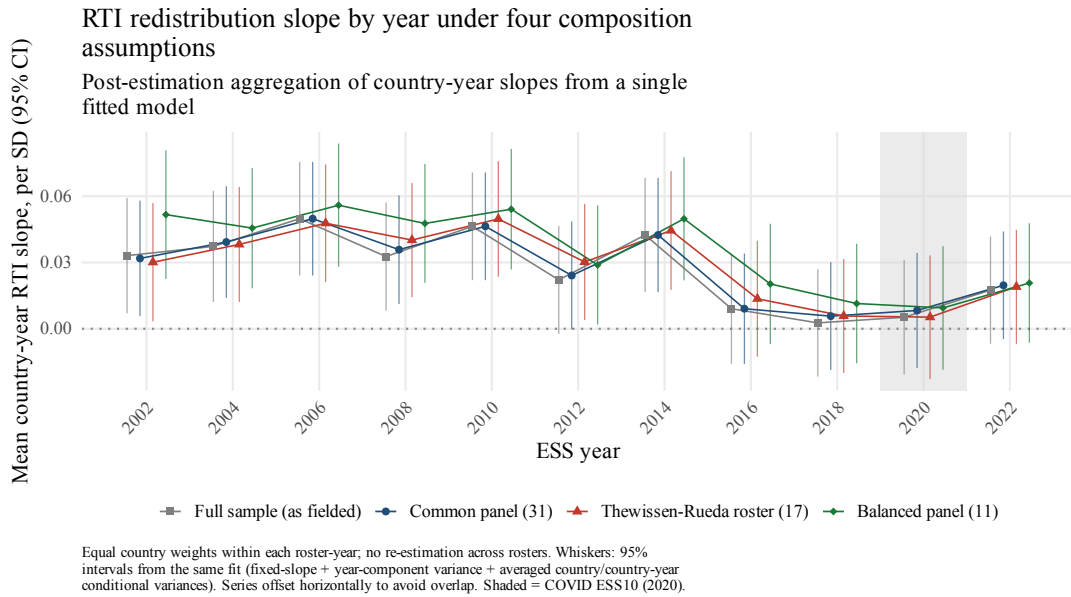


Table 4 lists the membership of each country roster used in the composition analysis, together with the number of observed waves per country.

Figure 13 plots design-weighted densities of both exposure indices by round for the seven-teen countries of the benchmark roster, with a common bandwidth per index. The distributions drift toward lower exposure across rounds. The design-weighted Frey–Osborne mean falls from

Table 4: Country membership in each analysis roster

Country	Waves	Full (39)	Common (31)	TR (17)	Balanced (11)
Albania (AL)	1	✓			
Austria (AT)	7	✓	✓	✓	
Belgium (BE)	11	✓	✓	✓	✓
Bulgaria (BG)	7	✓	✓		
Switzerland (CH)	11	✓	✓	✓	✓
Cyprus (CY)	6	✓	✓		
Czechia (CZ)	9	✓	✓		
Germany (DE)	10	✓	✓	✓	
Denmark (DK)	8	✓	✓	✓	
Estonia (EE)	10	✓	✓		
Spain (ES)	10	✓	✓	✓	
Finland (FI)	11	✓	✓	✓	✓
France (FR)	11	✓	✓	✓	✓
United Kingdom (GB)	11	✓	✓	✓	✓
Greece (GR)	6	✓	✓	✓	
Croatia (HR)	5	✓	✓		
Hungary (HU)	11	✓	✓		✓
Ireland (IE)	11	✓	✓	✓	✓
Israel (IL)	7	✓	✓		
Iceland (IS)	6	✓	✓		
Italy (IT)	6	✓	✓	✓	
Lithuania (LT)	7	✓	✓		
Luxembourg (LU)	2	✓		✓	
Latvia (LV)	3	✓	✓		
Montenegro (ME)	3	✓			
North Macedonia (MK)	1	✓			
Netherlands (NL)	11	✓	✓	✓	✓
Norway (NO)	11	✓	✓	✓	✓
Poland (PL)	10	✓	✓		
Portugal (PT)	11	✓	✓	✓	✓
Romania (RO)	1	✓			
Serbia (RS)	2	✓			
Russia (RU)	5	✓	✓		
Sweden (SE)	10	✓	✓	✓	
Slovenia (SI)	11	✓	✓		✓
Slovakia (SK)	8	✓	✓		
Turkey (TR)	2	✓			
Ukraine (UA)	6	✓	✓		
Kosovo (XK)	1	✓			

Waves counts the ESS rounds in which the country appears in the harmonized 2002–2022 panel (eleven rounds). Full: every country fielded in the panel. Common: at least three waves and at least one pre-2016 wave. TR: the seventeen countries of the Thewissen and Rueda [2019] main sample. Balanced: present in all eleven waves. Roster definitions follow the composition analysis code.

0.52 in 2002 to 0.47 in 2022, and the RTI mean falls from -0.04 to -0.20 over the same rounds, so the labor force of the benchmark roster moved away from routine-intensive occupations.

Figure 13: Design-weighted exposure distributions by ESS round, benchmark roster

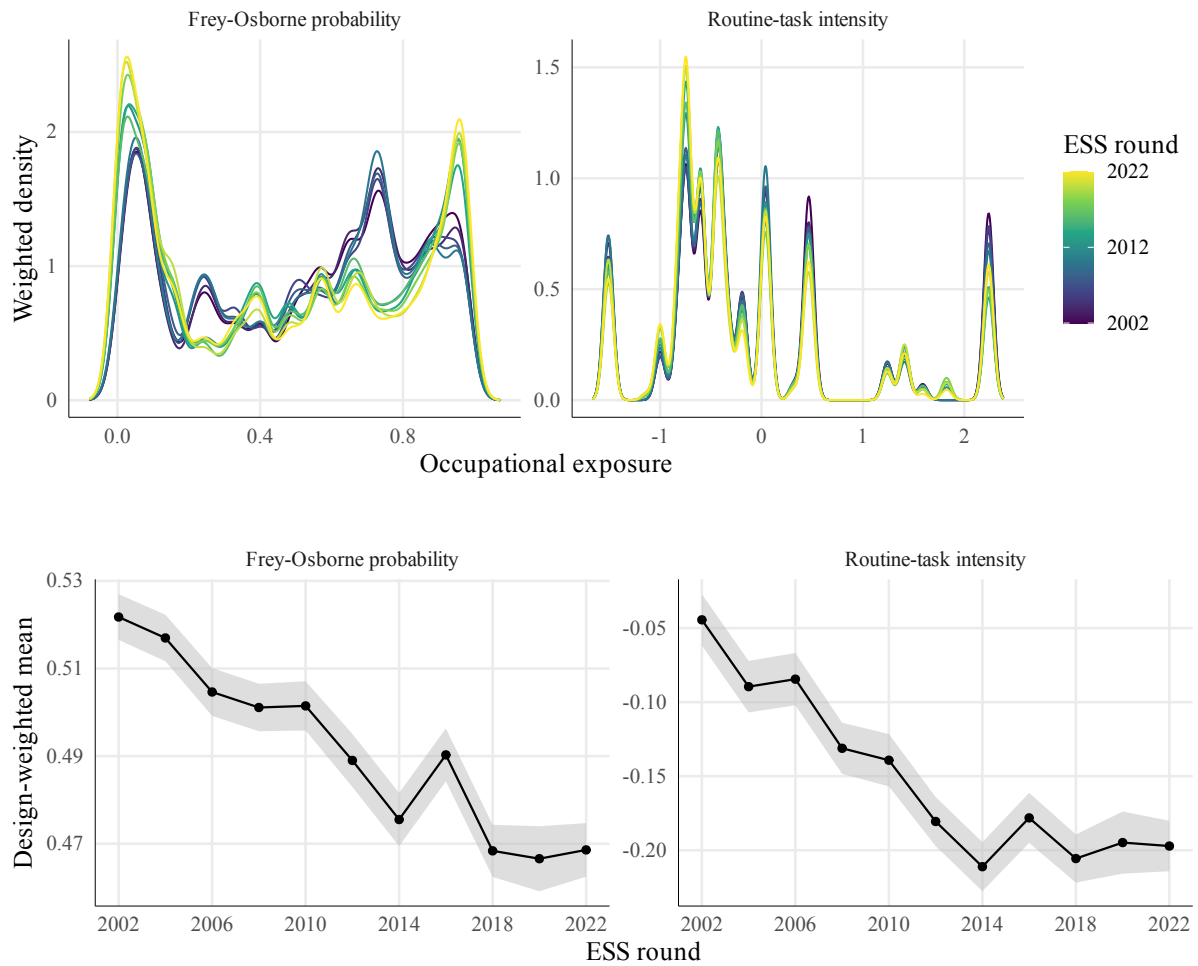
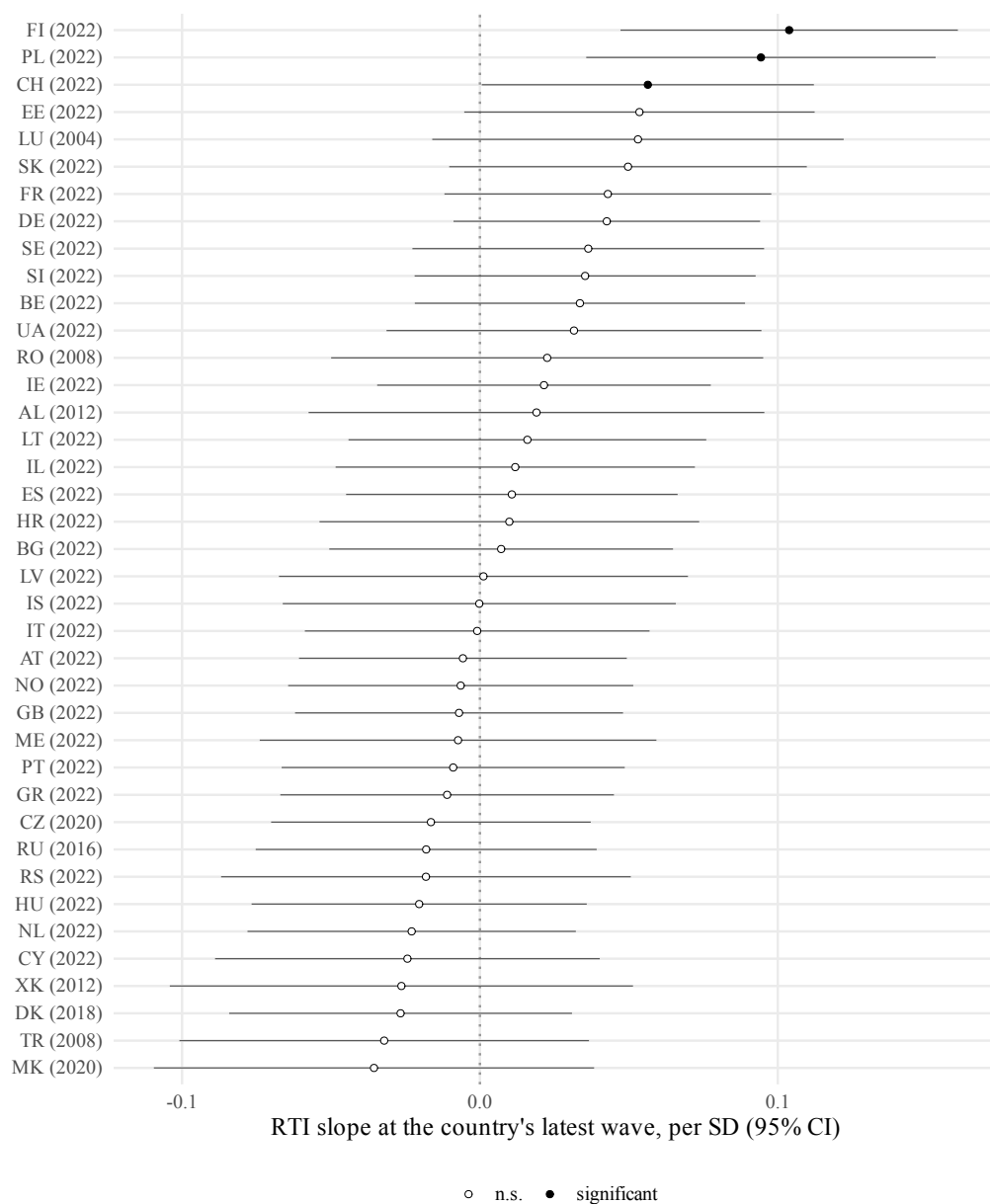


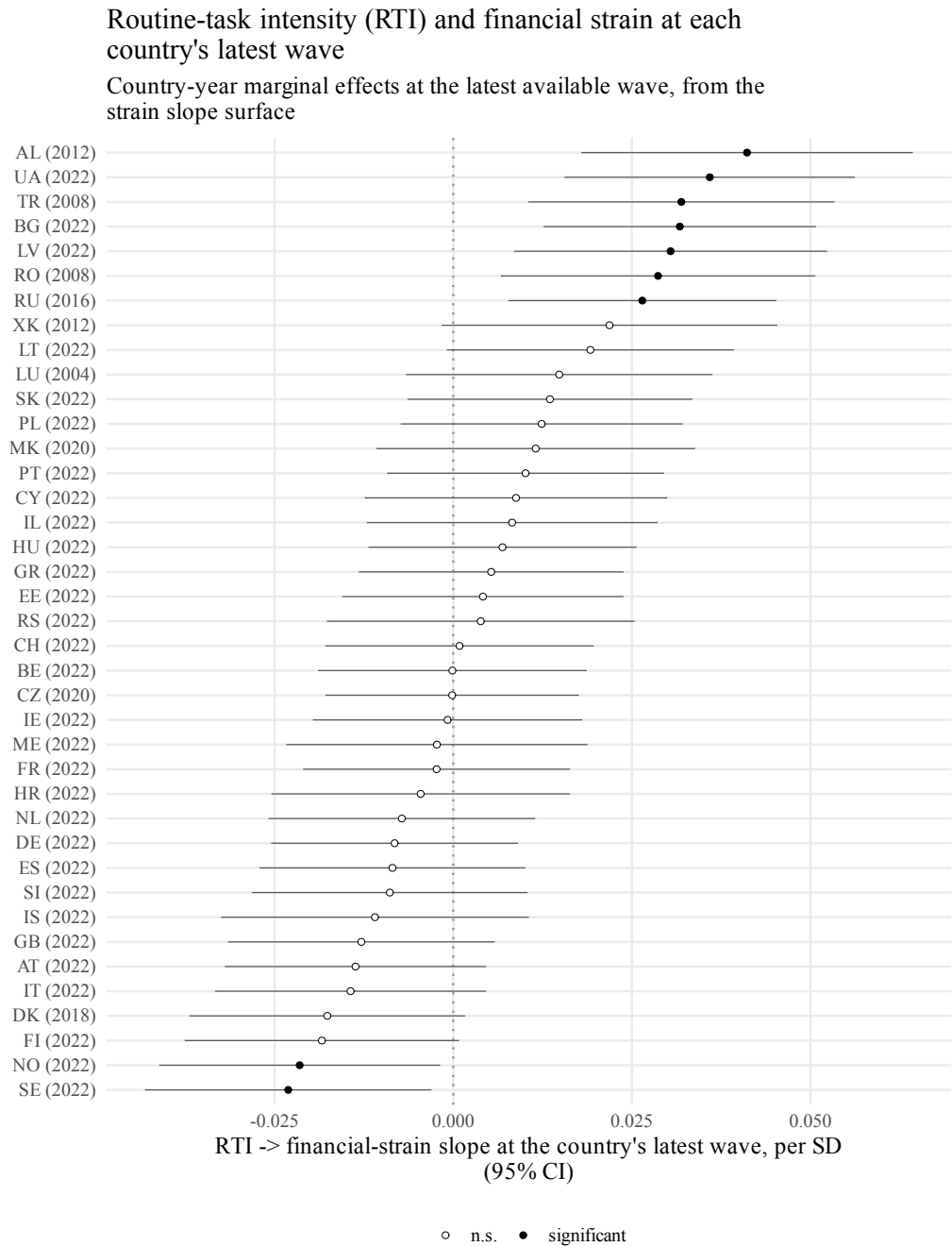
Figure 14: RTI robustness: country marginal effects at the most recent available wave

Routine-task intensity (RTI) and redistribution at each country's latest wave
Country-year marginal effects at the latest available wave, from the single model



Year in parentheses = the country's most recent ESS wave. Intervals combine the fixed-slope sampling variance with the conditional variances of the country, year, and country-year slope components. Dotted = 0.

Figure 15: RTI robustness: country marginal effects on financial strain at the most recent available wave

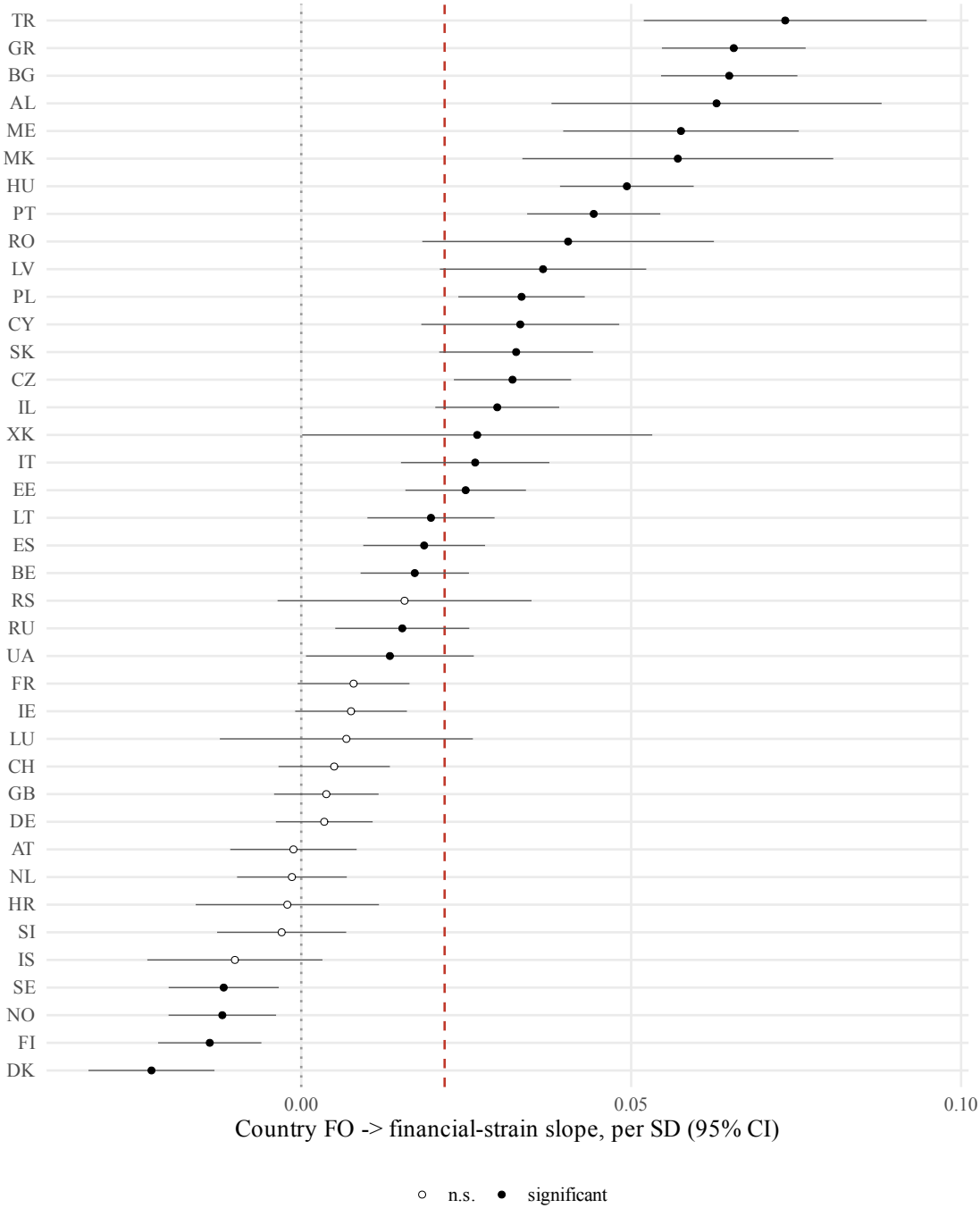


Year in parentheses = the country's most recent ESS wave. Intervals combine the fixed-slope sampling variance with the conditional variances of the country, year, and country-year slope components. Dotted = 0.

Figure 16: Country heterogeneity in automation exposure and financial strain, Frey–Osborne

Frey-Osborne (FO) and felt financial strain, by country (2002-2022)

Shared LPM (1 + x | country) + (1 + x | year); binary outcome:
difficult or very difficult on present income

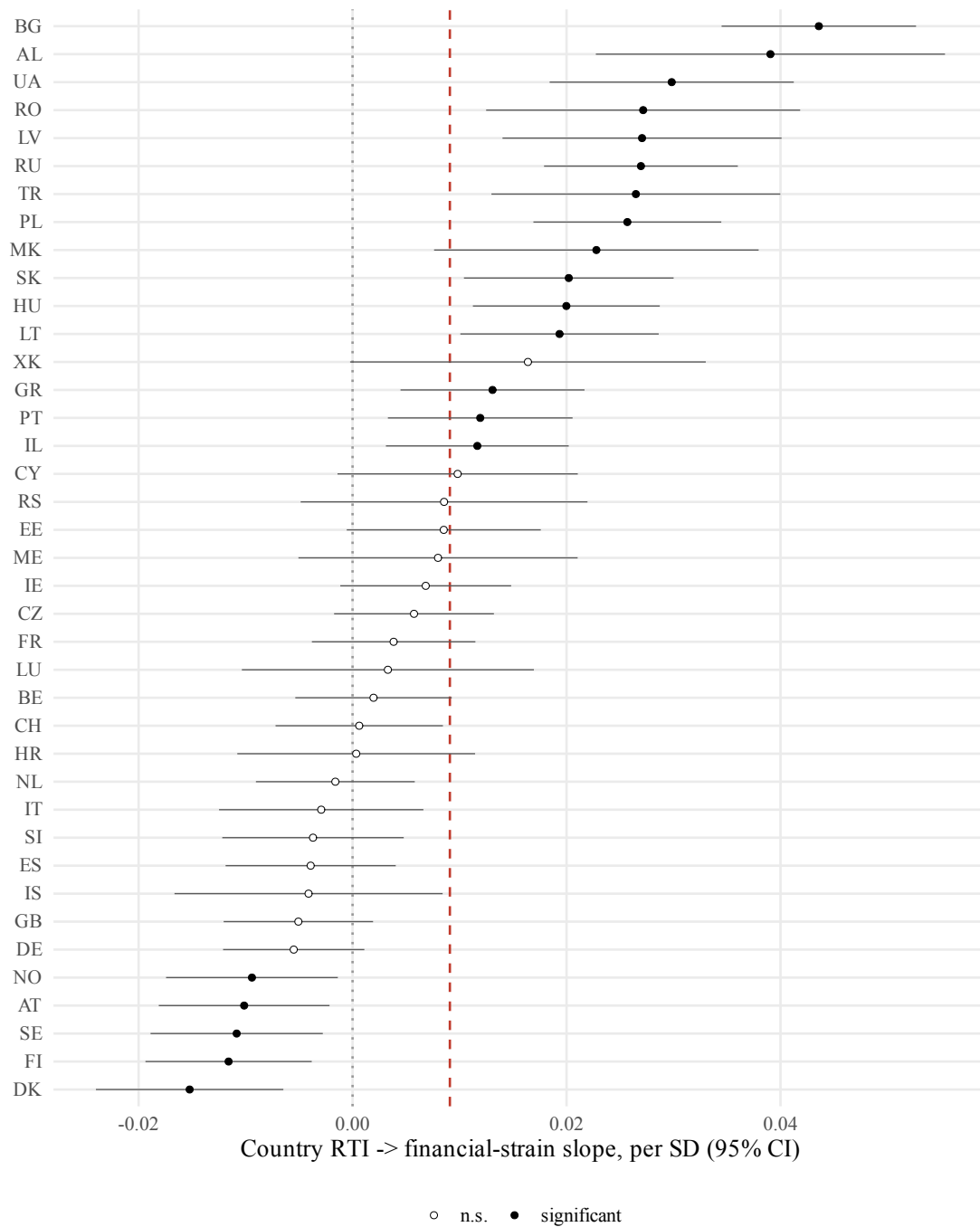


Dashed red = global slope (0.022); dotted = 0.

Figure 17: Country heterogeneity in automation exposure and financial strain, RTI

Routine-task intensity (RTI) and felt financial strain, by country (2002-2022)

Shared LPM (1 + x | country) + (1 + x | year); binary outcome:
difficult or very difficult on present income



Dashed red = global slope (0.009); dotted = 0.

Table 5: Mechanism diagnostic in the welfare-module rounds: automation risk and perceived economic vulnerability

Index	Job insecurity		Income insufficiency	
	2008	2016	2008	2016
Frey–Osborne probability	0.0319***	0.0232***	0.0238***	0.0187***
<i>N</i>	23637	21153	25616	21927
GMS routine-task intensity	0.0190***	0.0090*	0.0110**	0.0016
<i>N</i>	22266	19128	24169	19884

^a Each cell is the coefficient on the own-occupation automation-risk index in a design-weighted random-intercept linear probability model with country-clustered (CR1) standard errors. The exposure is standardized within the estimation sample, so a coefficient is the change in the probability of feeling vulnerable per one-standard-deviation increase in exposure. “Job insecurity” is a binary indicator for expecting to lose one’s job in the next twelve months; “income insufficiency” for expecting income too low to meet needs. The welfare-attitudes module carries both items in the 2008 and 2016 rounds; the job models use the paid-work frame and the income models the labor-force frame. Positive coefficients mean higher exposure predicts greater perceived vulnerability.

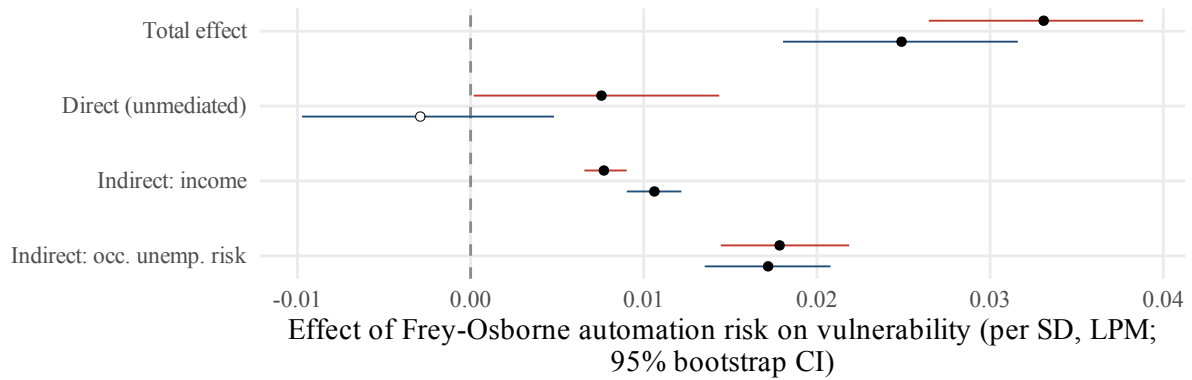
^b Stars are Holm-adjusted within each outcome-wave across the two indices: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The confirmatory registration covers the 2016 round; the 2008 columns are an out-of-registration extension under the same specification, with occupational exposure taken from the panel’s ISCO-88-harmonized indices. Country counts differ across waves and outcomes through module coverage and listwise deletion.

^c A missing-code recode in the registered pipeline also removed respondents aged 77, 88, and 99; the reported estimates correct this and otherwise preserve the registered specification.

Figure 18: Mediation decomposition and path diagrams (2016 wave)

How Frey-Osborne automation risk translates into perceived vulnerability (ESS8 2016)

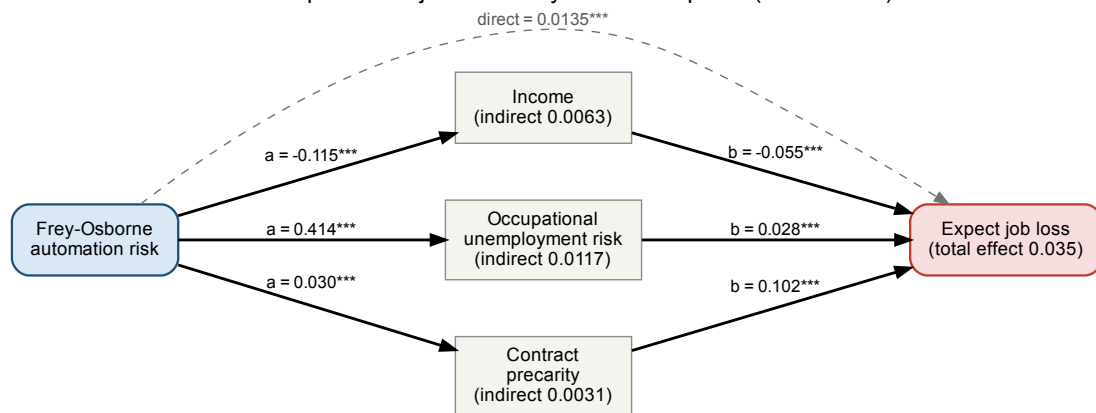
Parallel multiple-mediator decomposition; minimal backdoor (educ, age, sex) + country FE



○ n.s. ● significant — Expect job loss — Expect low income

Total = direct + indirect(income) + indirect(occupational unemployment risk). Mediators modelled as paths, not controls.

FO and perceived job insecurity: mediation paths (ESS8 2016)



FO and perceived income insufficiency: mediation paths (ESS8 2016)

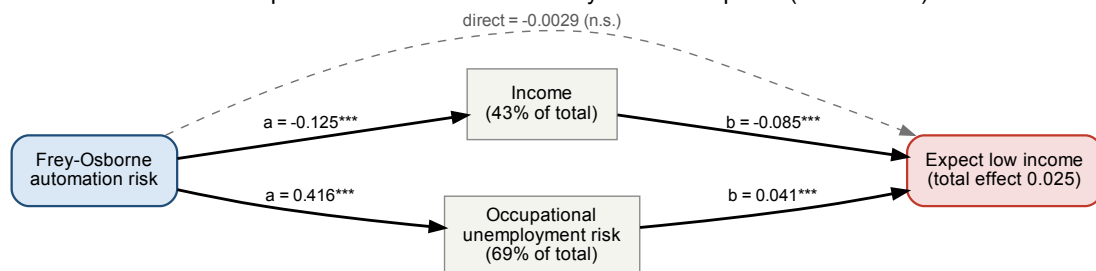
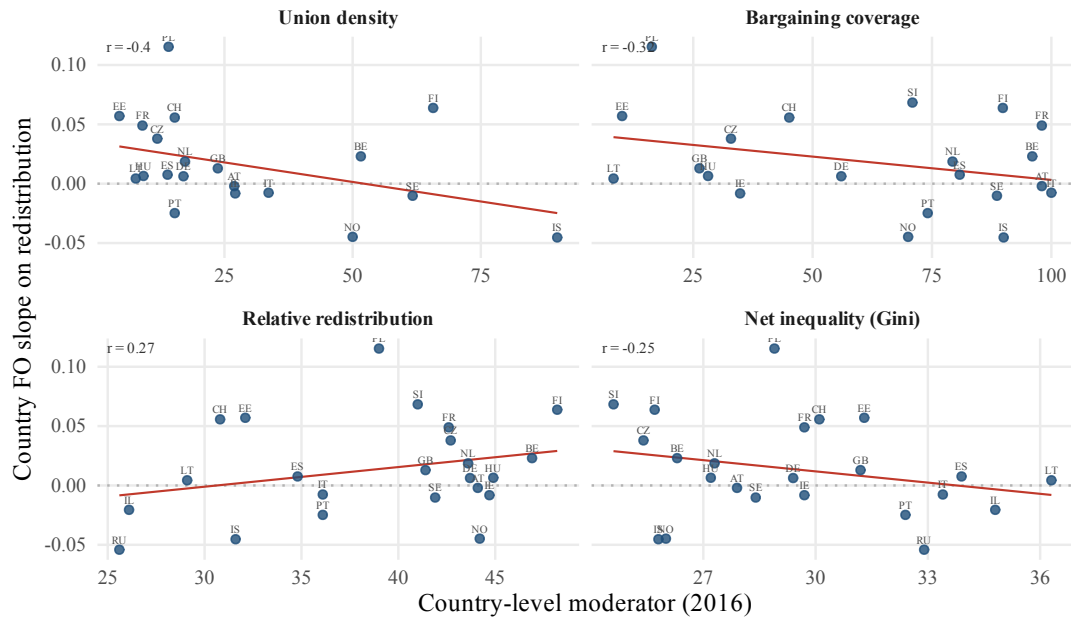


Figure 19: Two-stage institutional diagnostic for redistribution heterogeneity

Two-stage view: country FO slope vs the strongest institutional moderators

Each point a country (2016 wave); line = OLS across countries



beta_c = beta_{c,2016} (FO per SD) from the single heterogeneity model.

Table 6: RTI robustness: between- and within-country institutional moderation of the automation–redistribution gradient (Mundlak decomposition)

Moderator	Between country		Within country		<i>N</i>	Countries
	Est.	(SE)	Est.	(SE)		
Net inequality (Gini)	-0.0147*	(0.0059)	-0.0037	(0.0026)	179,577	38
Social spending %GDP	+0.0083	(0.0061)	-0.0031	(0.0026)	169,413	31
Relative redistribution	+0.0147	(0.0062)	+0.0001	(0.0026)	174,922	33
UB net replacement rate	+0.0079	(0.0056)	+0.0025	(0.0027)	166,004	31
Left cabinet share	+0.0045	(0.0062)	+0.0035	(0.0024)	165,600	30
Union density	-0.0026	(0.0064)	+0.0121**	(0.0032)	139,823	30
Bargaining coverage	+0.0013	(0.0064)	+0.0159*	(0.0063)	139,641	30
Employment protection (EPL)	-0.0029	(0.0062)	-0.0068	(0.0029)	135,299	25
Industry employment share	+0.0032	(0.0058)	+0.0017	(0.0025)	165,600	30
Trade openness	-0.0007	(0.0068)	-0.0107**	(0.0033)	149,077	30
Benefit earnings-relatedness	+0.0005	(0.0059)	-0.0050	(0.0025)	166,004	31

Each row is one multilevel model of support for redistribution on standardized routine-task-intensity (RTI) exposure, following the Mundlak (hybrid) specification: the moderator, lagged one survey year, is split into its country mean (between component) and the deviation from that mean (within component), both standardized, and both interacted with exposure in the same model. Cells report the cross-level interaction coefficients. All models include exposure interactions with contemporaneous unemployment and real GDP growth, the baseline individual controls, year fixed effects, a random exposure slope by country, and a country-year random intercept, with ESS design weights and maximum likelihood estimation. Stars mark Benjamini–Hochberg adjusted *p* values, corrected across the full family of 44 coefficients (eleven moderators, two components, two exposure indices): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

8.3 NACE Rev. 2.1: First-Level Economic-Activity Classification

Table 7: Statistical Classification of Economic Activities in the European Community, Rev. 2.1 (NACE Rev. 2.1)

Section	Code Range	Description
A	01:03	Agriculture, Forestry and Fishing
B	05:09	Mining and Quarrying
C	10:33	Manufacturing
D	35:35	Electricity, Gas, Steam and Air Conditioning Supply
E	36:39	Water Supply; Sewerage, Waste Management and Remediation Activities
F	41:43	Construction
G	45:47	Wholesale and Retail Trade
H	49:53	Transportation and Storage
I	55:56	Accommodation and Food Service Activities
J	58:60	Publishing, Broadcasting, and Content Production and Distribution Activities
K	61:63	Telecommunication, Computer Programming, Consulting, Computing Infrastructure and Other Information Service Activities
L	64:66	Financial and Insurance Activities
M	68:68	Real Estate Activities
N	69:75	Professional, Scientific and Technical Activities
O	77:82	Administrative and Support Service Activities
P	84:84	Public Administration and Defence; Compulsory Social Security
Q	85:85	Education
R	86:88	Human Health and Social Work Activities
S	90:93	Arts, Sports and Recreation
T	94:96	Other Service Activities
U	97:98	Activities of Households as Employers and Undifferentiated Goods - and Service-Producing Activities of Households for Own Use
V	99:99	Activities of Extraterritorial Organisations and Bodies

8.4 Standard Occupational Classification (SOC) System

The 2010 Standard Occupational Classification (SOC-2010) is the U.S. statistical system for classifying occupations. Federal agencies use it to organize occupational information and labor-market data.

SOC-2010 is hierarchical, ranging from broad groups to specific occupations identified by unique codes. It contains more than 800 detailed occupation codes.

Major Categories of SOC-2010

The table below lists the SOC-2010 major groups.

Table 8: Major Categories of SOC-2010

Code	Major Group	Code	Major Group
11	Management Occupations	33	Protective Service Occupations
13	Business and Financial Operations Occupations	35	Food Preparation and Serving Related Occupations
15	Computer and Mathematical Occupations	37	Building and Grounds Cleaning and Maintenance Occupations
17	Architecture and Engineering Occupations	39	Personal Care and Service Occupations
19	Life, Physical, and Social Science Occupations	41	Sales and Related Occupations
21	Community and Social Service Occupations	43	Office and Administrative Support Occupations
23	Legal Occupations	45	Farming, Fishing, and Forestry Occupations
25	Education, Training, and Library Occupations	47	Construction and Extraction Occupations
27	Arts, Design, Entertainment, Sports, and Media Occupations	49	Installation, Maintenance, and Repair Occupations
29	Healthcare Practitioners and Technical Occupations	51	Production Occupations
31	Healthcare Support Occupations	53	Transportation and Material Moving Occupations

8.5 Descriptive Statistics

Table 9: Descriptive statistics for the analysis variables of the harmonised 2002–2022 labour-force panel (all eleven ESS waves). Each row is a model variable; the columns report the number of non-missing observations (N), the mean and standard deviation, and the minimum, the tenth, twenty-fifth, fiftieth, seventy-fifth, and ninetieth percentiles, and the maximum. Binary indicators have means equal to their prevalence in the sample. The Frey–Osborne automation probability is on its native 0–1 scale, the routine-task-intensity index on its native GMS scale, and relative equivalized income enters as the within-country-year z-score. The vulnerability items exist only in the waves noted.

Variable	N	Mean	SD	Min	p10	p25	Median	p75	p90	Max
Redistribution support (1–5) (<i>y</i>)	271,513	3.84	1.05	1.00	2.00	3.00	4.00	5.00	5.00	5.00
Frey–Osborne automation probability (<i>fo</i>)	256,851	0.51	0.34	0.00	0.04	0.13	0.57	0.81	0.95	0.99
Routine-task intensity (GMS) (<i>rti</i>)	237,440	-0.14	0.95	-1.52	-1.00	-0.73	-0.44	0.32	1.41	2.24
Relative equivalized income (<i>z</i>) (<i>income</i>)	215,274	0.21	1.06	-21.31	-0.79	-0.45	-0.02	0.60	1.54	24.39
Years of education (<i>edyrs</i>)	272,624	13.60	3.74	0.00	9.00	12.00	13.00	16.00	18.00	56.00
Age (<i>age</i>)	273,697	42.75	12.27	14.00	26.00	33.00	43.00	52.00	59.00	114.00
Male (<i>male</i>)	274,864	0.52	0.50	0.00	0.00	0.00	1.00	1.00	1.00	1.00
Union member (current/former) (<i>union</i>)	272,469	0.41	0.49	0.00	0.00	0.00	0.00	1.00	1.00	1.00
Religiosity (0–10) (<i>relig</i>)	272,509	4.34	2.98	0.00	0.00	2.00	5.00	7.00	8.00	10.00
Unemployed (active) (<i>unemp</i>)	274,988	0.07	0.26	0.00	0.00	0.00	0.00	0.00	0.00	1.00
Financial strain (difficult on income) (<i>d_fin_strain</i>)	270,942	0.22	0.41	0.00	0.00	0.00	0.00	0.00	1.00	1.00
Expects job loss (2008, 2016) (<i>d_job_vulnr</i>)	51,606	0.26	0.44	0.00	0.00	0.00	0.00	1.00	1.00	1.00
Expects insufficient income (2008, 2016) (<i>d_income_vulnr</i>)	51,500	0.29	0.45	0.00	0.00	0.00	0.00	1.00	1.00	1.00
Job not secure (2004, 2010) (<i>d_job_insecure</i>)	37,982	0.36	0.48	0.00	0.00	0.00	0.00	1.00	1.00	1.00
Occupational unemployment risk (<i>occ_risk</i>)	266,238	0.06	0.05	0.00	0.02	0.03	0.05	0.08	0.12	1.00
Temporary or no contract (<i>no_or_limtd_contract</i>)	215,002	0.21	0.41	0.00	0.00	0.00	0.00	0.00	1.00	1.00

8.6 ESS Variable Dictionary

Table 10: ESS variable dictionary

Variable	Description	Value labels
y	Redistribution preference; the ESS item gincdif reverse-coded ($y = 6 - \text{gincdif}$) so higher values mean stronger support for the government reducing income differences.	'5' = 'Agree strongly'; '4' = 'Agree'; '3' = 'Neither agree nor disagree'; '2' = 'Disagree'; '1' = 'Disagree strongly'
fo	Frey-Osborne (2017) probability that the respondent's occupation is automatable, matched to the occupation code and harmonized across the ISCO-88 and ISCO-08 waves.	Continuous, 0 to 1
rti	Goos-Manning-Salomons routine-task-intensity of the occupation (appendix robustness exposure), harmonized across the ISCO seam.	Continuous; higher is more routine-task-intensive
income	Thewissen-Rueda relative equivalized household income, entered as the within country-year z-score. Built from the ESS show-card income band midpoint (Pareto mean for the open top band), square-root household equivalization, and the ratio to the design-weighted country-year mean.	Continuous z-score
eduyrs	Years of full-time education completed (ESS eduyrs).	Continuous, years
male	Respondent gender, from ESS gndr.	'1' = 'Male'; '0' = 'Female'
age	Age of the respondent in years (ESS agea).	Continuous, years
union	Trade-union membership, from ESS mbtru.	'1' = 'Member, current or former'; '0' = 'Never a member'
relig	Subjective religiosity on the ESS rlgdgr scale.	'0' = 'Not at all religious'; '10' = 'Very religious'
unemp	Unemployed under the Thewissen-Rueda labour-force definition (ESS mnactic coded as unemployed, looking or not looking for work).	'1' = 'Unemployed'; '0' = 'Employed'

Continued on next page

Variable	Description	Value labels
occ_risk	Occupational unemployment risk, Rehm-style: the design-weighted, leave-one-out unemployment rate by wave and two-digit ISCO under the ILO labour-force definition (minimum cell size of 30).	Continuous, share unemployed
no_or_limtd_contract	Contract precarity, from ESS wrkctra.	'1' = 'Limited-duration or no contract'; '0' = 'Unlimited contract'
d_fin_strain	Subjective financial strain, from ESS hincfel; available in all eleven waves.	'1' = 'Difficult or very difficult on present income'; '0' = 'Coping or living comfortably'
d_job_vulnr	Expected job loss, from ESS lkuemp (likely or very likely to become unemployed in the next twelve months). Welfare module, 2008 and 2016.	'1' = 'Likely or very likely'; '0' = 'Not or not very likely'
d_income_vulnr	Expected income insufficiency, from ESS lknemny (likely or very likely to have insufficient household income in the next twelve months). Welfare module, 2008 and 2016.	'1' = 'Likely or very likely'; '0' = 'Not or not very likely'
d_job_insecure	Job insecurity, from ESS jbscr (rates own job security low). Work module, 2004 and 2010.	'1' = 'Insecure'; '0' = 'Secure'
dweight	ESS design weight, applied in all estimation.	Continuous weight
country	ESS country of interview (39 countries, ISO codes).	Categorical
year	ESS survey year and wave, 2002 to 2022 (rounds 1 to 11).	Categorical, 2002 to 2022

8.7 Institutional Measures Dictionary

Table 11 defines each institutional moderator and records its source series and country coverage.

Table 11: Institutional moderators: definitions and sources

Variable	Measure	Description	Source	Ctries
gininet	Net inequality (Gini)	Gini index of net (post-tax, post-transfer) income inequality, in Gini points.	SWIID, version 9.92	38
relred	Relative redistribution	Reduction of market-income inequality achieved by taxes and transfers, as recorded in the SWIID relative-redistribution series (percent).	SWIID, version 9.92	33
socgdp	Social spending %GDP	Public social expenditure as a share of GDP (percent).	OECD SOCX	31
nrr_m6	UB net replacement rate	Net unemployment-benefit replacement rate for a single person without children at 67% of the average wage, six months into unemployment, without social-assistance top-ups (percent).	OECD tax-benefit models	31
nrr_decay_m6	Benefit earnings-relatedness	Decay of the net replacement rate as previous earnings rise: NRR at 100% of the average wage minus NRR at 67%, six months into unemployment (percentage points). Values near zero indicate earnings-related benefits; strongly negative values indicate flat or capped benefits.	OECD tax-benefit models	31
ud_ipol	Union density	Union density, interpolated from the OECD/AIAS ICTWSS series (percent of employees).	CPDS (OECD/AIAS ICTWSS)	30
adjcov_ipol	Bargaining coverage	Adjusted collective-bargaining coverage, interpolated from the OECD/AIAS ICTWSS series (percent of employees).	CPDS (OECD/AIAS ICTWSS)	30
emprot_reg	Employment protection (EPL)	Strictness of employment protection for regular contracts (index), per the CPDS series emprot_reg.	CPDS	25
gov_left1	Left cabinet share	Share of cabinet posts held by left parties (percent), per the CPDS series gov_left1.	CPDS	30
emp_ind	Industry employment share	Employment in industry, per the CPDS series emp_ind.	CPDS	30
openc	Trade openness	Trade openness: exports plus imports as a share of GDP (percent), per the CPDS series openc.	CPDS	30

Ctries: national clusters available to the moderation models, from the estimation output. Moderators enter the models standardized and lagged one survey year. CPDS: Comparative Political Data Set [Armingeon et al., 2025]. SWIID: Standardized World Income Inequality Database.